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OS04J10

datasheet

PRELIMINARY SPECIFICATION

1/1.88" color CMOS 4 megapixel image sensor

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color CMOS 4 megapixel image sensor

datasheet (CSP)
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applications

- security surveillance systems
- IP cameras
- HD analog cameras

ordering information

- **OS04J10-A70A-001A-Z** (color, lead-free)
70-pin CSP
- **OS04J10-A70A-001B-Z** (color, lead-free)
70-pin CSP

features

- programmable controls: frame rate, mirror and flip, cropping, and windowing
- supports 2x2 color and monochrome binning function
- support for output formats: 10-bit 4-lane MIPI TX
- supports image sizes: full size (2560 x 1440), 1080p (1920 x 1080), and 720p (1280 x 720)
- SCCB control interface for register programming
- supports MIPI serial output interface (2-lane/4-lane)
- supports 2K bits of one-time programmable (OTP) memory
- supports dynamic and OTP defective pixel correction (DPC)
- supports automatic black level calibration (BLC)
- supports multi-camera synchronous function

key specifications (typical)

- **active array size:** 2560 x 1440
- **power supply:**
 - analog: 2.8V
 - I/O: 1.8V
 - core: 1.2V
- **power requirements:** (see **sidebar note**)
 - active: <200 mW
 - standby: <1 mA
- **temperature range:**
 - operating: -30°C to +85°C junction temperature (see **table 6-2**)
 - stable image: 0°C to +60°C junction temperature (see **table 6-2**)
- **output interfaces:** 10-bit 4-lane MIPI
- **output formats:**
 - linear: 10/12-bit RGB RAW
 - 2-exposure staggered HDR: 10-bit RGB RAW
 - DCG combine HDR: 12/16-bit RGB RAW
- **lens size:** 1/1.88"
- **lens chief ray angle:** 12° linear (see **figure 8-1**)
- **input clock frequency:** 6~27 MHz
- **maximum image transfer rate:**
 - linear mode:
 - full size 2560 x 1440 @ 60 fps @ 10-bit
 - full size 2560 x 1440 @ 30 fps @ 12-bit
 - 2-exposure staggered HDR mode:
 - full size 2560 x 1440 @ 30 fps x2 @ 10-bit
 - DCG combine HDR mode:
 - full size 2560 x 1440 @ 30 fps @ 10-bit
- **sensitivity:** 31661 e⁻/Lux-sec
- **shutter:** rolling
- **max S/N ratio:** 35.4 dB @ HCG, 42.2 dB @ LCG
- **dynamic range:** 66.2 dB @ 1x, 77.6 dB @ 15.5x
- **pixel size:** 2.898 μm x 2.898 μm
- **image area:** 7442.064 μm x 4196.304 μm
- **package dimensions:** 8342.4 μm x 5214.9 μm (see **section 7**)



note Power requirements specifications are estimated values and are subject to change when measured data using real silicon is available.

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1 signal descriptions

table 1-1 lists the signal descriptions and their corresponding pin numbers for the OS04J10 image sensor. The package information is shown in **section 7**.

figure 1-1 pin diagram

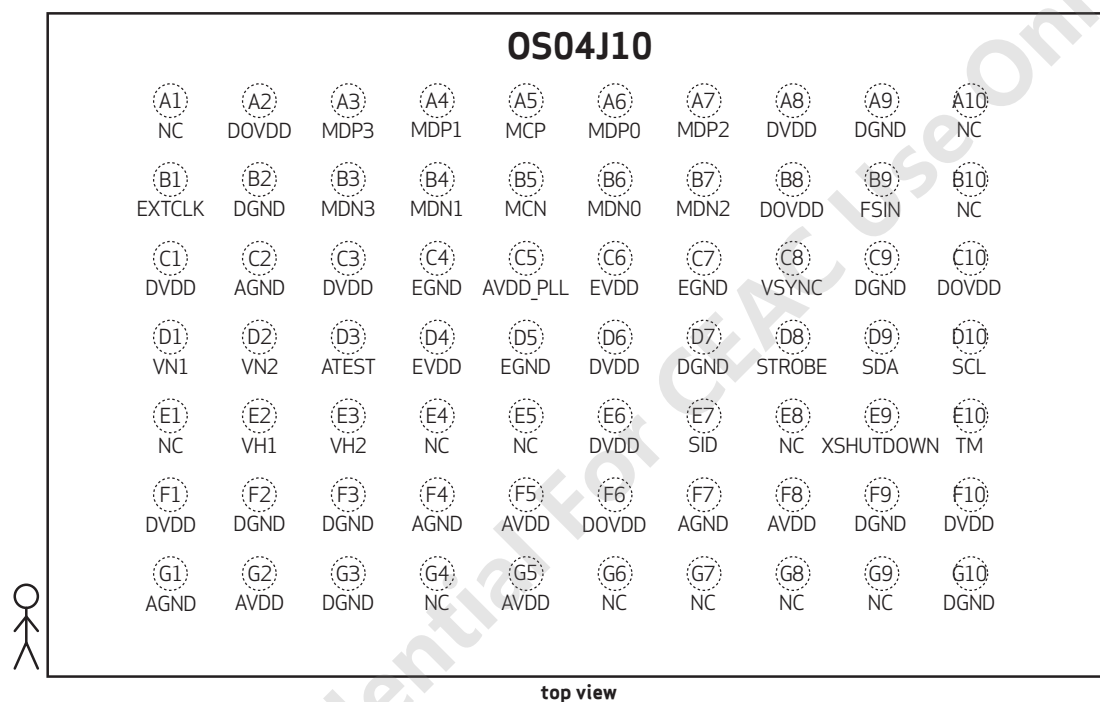


table 1-1 signal descriptions (sheet 1 of 4)

pin number	signal name	pin type	description
A1	NC	—	no connect
A2	DOVDD	power	I/O power 1.8V
A3	MDP3	output	MIPI data positive output 3
A4	MDP1	output	MIPI data positive output 1
A5	MCP	output	MIPI clock positive output
A6	MDP0	output	MIPI data positive output 0
A7	MDP2	output	MIPI data positive output 2

table 1-1 signal descriptions (sheet 2 of 4)

pin number	signal name	pin type	description
A8	DVDD	power	digital circuit power 1.2V
A9	DGND	ground	ground for I/O and digital circuit
A10	NC	—	no connect
B1	EXTCLK	input	clock input
B2	DGND	ground	ground for I/O and digital circuit
B3	MDN3	output	MIPI data negative output 3
B4	MDN1	output	MIPI data negative output 1
B5	MCN	output	MIPI clock negative output
B6	MDN0	output	MIPI data negative output 0
B7	MDN2	output	MIPI data negative output 2
B8	DOVDD	power	I/O power 1.8V
B9	FSIN	input	frame synchronization input
B10	NC	—	no connect
C1	DVDD	power	digital circuit power 1.2V
C2	AGND	ground	analog ground
C3	DVDD	power	digital circuit power
C4	EGND	ground	MIPI supply ground
C5	AVDD_PLL	power	PLL power supply
C6	EVDD	power	MIPI core logic power
C7	EGND	ground	MIPI supply ground
C8	VSNC	output	vertical/frame synchronization output signal
C9	DGND	ground	ground for I/O and digital circuit
C10	DOVDD	power	I/O power 1.8V
D1	VN1	power	negative reference voltage 1 (connect to 1 μ F capacitor outside)
D2	VN2	power	negative reference voltage 2 (connect to 1 μ F capacitor outside)
D3	ATEST	analog I/O	analog test I/O
D4	EVDD	power	MIPI core logic power
D5	EGND	ground	MIPI supply ground
D6	DVDD	power	digital circuit power 1.2V
D7	DGND	ground	ground for I/O and digital circuit

table 1-1 signal descriptions (sheet 3 of 4)

pin number	signal name	pin type	description
D8	STROBE	output	frame exposure output indicator / exposure synchronization input
D9	SDA	I/O	SCCB interface data pin
D10	SCL	input	SCCB interface input clock
E1	NC	–	no connect
E2	VH1	power	positive reference voltage 1 (connect to 1μF capacitor outside)
E3	VH2	power	positive reference voltage 2 (connect to 1μF capacitor outside)
E4	NC	–	no connect
E5	NC	–	no connect
E6	DVDD	power	digital circuit power 1.2V
E7	SID	input	sensor SCCB ID switch input 0: SCCB ID address = 0x78 1: SCCB ID address = 0x7A
E8	NC	–	no connect
E9	XSHUTDOWN	input	reset and power down (active low with pull-down resistor)
E10	TM	input	test mode (active high, internally tied to pull down resistor)
F1	DVDD	power	digital circuit power 1.2V
F2	DGND	ground	ground for I/O and digital circuit
F3	DGND	ground	ground for I/O and digital circuit
F4	AGND	ground	analog ground
F5	AVDD	power	analog power 2.8V
F6	DOVDD	power	I/O power 1.8V
F7	AGND	ground	analog ground
F8	AVDD	power	analog power 2.8V
F9	DGND	ground	ground for I/O and digital circuit
F10	DVDD	power	digital circuit power 1.2V
G1	AGND	ground	analog ground
G2	AVDD	power	analog power 2.8V
G3	DGND	ground	ground for I/O and digital circuit
G4	NC	–	no connect
G5	AVDD	power	analog power 2.8V
G6	NC	–	no connect

table 1-1 signal descriptions (sheet 4 of 4)

pin number	signal name	pin type	description
G7	NC	–	no connect
G8	NC	–	no connect
G9	NC	–	no connect
G10	DGND	ground	ground for I/O and digital circuit

table 1-2 pin states under various conditions

pin	signal name	XSHUTDOWN = 0	XSHUTDOWN = 1	software standby
A3	MDP3	high-z	high	high
A4	MDP1	high-z	high	high
A5	MCP	high-z	high	high
A6	MDP0	high-z	high	high
A7	MDP2	high-z	high	high
B1	EXTCLK	input	input	input
B3	MDN3	high-z	high	high
B4	MDN1	high-z	high	high
B5	MCN	high-z	high	high
B6	MDN0	high-z	high	high
B7	MDN2	high-z	high	high
B9	FSIN	high-z	low	input (configurable)
C8	VSYNC	high-z	low	input ^a (configurable)
D3	ATEST	high-z	high-z	high-z
D8	STROBE	low	low	low
D9	SDA	open drain	open drain	I/O
D10	SCL	high-z	high-z	input
E7	SID	input	input	input
E9	XSHUTDOWN	input	input	input
E10	TM	input	input	input

a. default state can change, depending on state of sensor and/or function of pin

table 1-3 pad equivalent circuit (sheet 1 of 2)

signal name	equivalent circuit
EXTCLK	
SDA	
SID	
XSHUTDOWN, TM	
SCL	
FSIN, STROBE, VSYNC	
MCN, MCP, MDN[3:0], MDP[3:0]	
AGND, ATEST, DGND, EGND, VH1, VH2	

table 1-3 pad equivalent circuit (sheet 2 of 2)

signal name	equivalent circuit
VN1, VN2	
AVDD, AVDD_PLL, DVDD, EVDD, DOVDD	

2 system level description

2.1 overview

The OS04J10 image sensor is a high performance, 1/1.88 inch, 4 megapixel, CMOS image sensor that provides up to 2560x1440 video at 60 fps. It provides a full size or windowed frame MIPI interface, in various formats, for control over via serial camera control bus (SCCB) interface. The OS04J10 focuses on products including security surveillance systems, IP cameras and HD analog cameras.

By introducing an advanced 2.9 μm pixel architecture, the OS04J10 achieves excellent low-light sensitivity, signal-to-noise ratio (SNR), full-well capacity, quantum efficiency and low-power consumption. The default and programmable modes allow for a more convenient method of controlling the parameters of frame size, exposure time, gain value, etc. It also offers the following image control functions: mirror and flip, windowing, auto black level calibration, defective pixel correction, dual conversion gain (DCG) combine, and other functions.

The OS04J10 supports a high frame rate, up to 60 fps, 10/12/16-bit MIPI interface, with linear, HDR, DCG, and DCG combine modes. These prominent features integrated in the OS04J10 allow for a best-in-class image sensor that will bring users vivid pictures and an excellent experience.

2.2 architecture

The OS04J10 sensor core generates streaming pixel data at a constant frame rate. **figure 2-1** shows the functional block diagram of the OS04J10 image sensor.

The timing generator outputs clocks to access the rows of the imaging array, pre-charging and sampling the rows of the array sequentially. In the time between pre-charging and sampling a row, the charge in the pixels decreases with exposure to incident light. This is the exposure time in rolling shutter architecture.

The exposure time is controlled by adjusting the time interval between pre-charging and sampling. After the data of the pixels in the row has been sampled, it is processed through analog circuitry to correct the offset and multiply the data with corresponding gain. Following analog processing is the ADC, which outputs up to 12 bits of data for each pixel in the array.

figure 2-1 OS04J10 block diagram

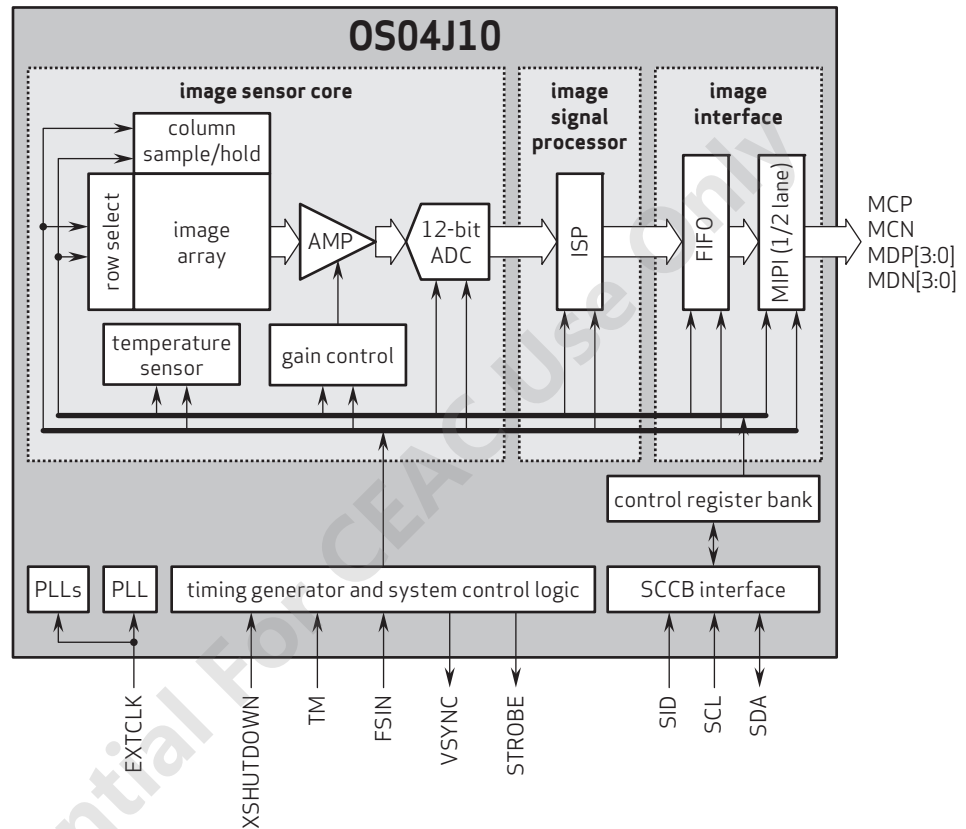
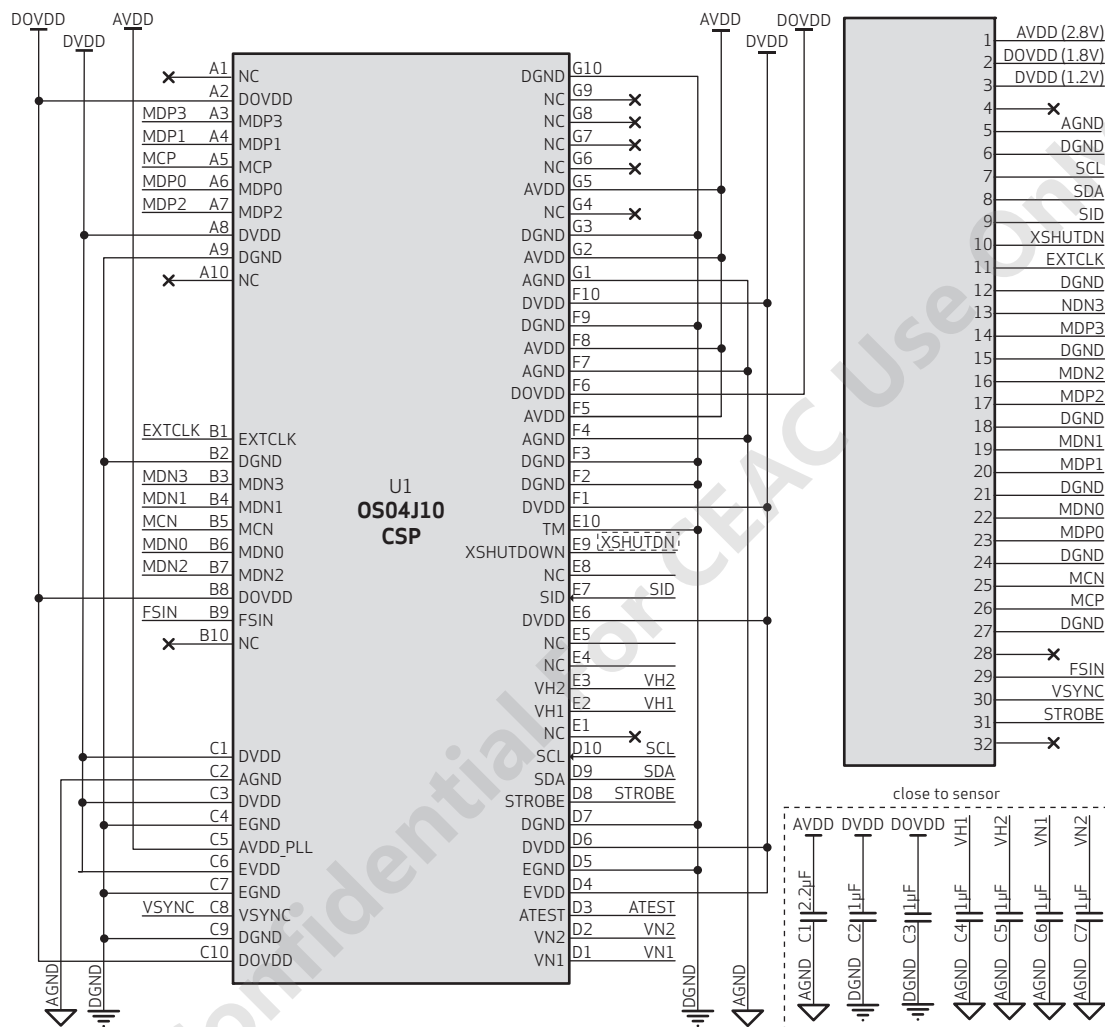


figure 2-2 OS04J10 reference schematic



- note 1** client platform provides three power supplies for AVDD, DOVDD, and DVDD; AVDD is connected to 2.8V power supply, DOVDD is compatible with 1.8V power supply, and DVDD is connected to 1.2V power supply; VH1, VH2, VN1, and VN2 are all connected with a 1 μ F capacitor and then grounded, without leading out of interface; LDO type: WL2836Exx series
- note 2** XSHUTDOWN(XSHUTDN) must be controlled by platform GPIO
- note 3** this design is MIPI mode; SID is SCCB address controllable pin; when SID is connected low, write address is 0x78 and read address is 0x79; when SID is connected high, write address is 0x7A and read address is 0x7B
- note 4** in PCB design, place chip power filter capacitors close to power supply pins; place a 2.2 μ F capacitor near AVDD and a 1 μ F capacitor near DOVDD, DVDD, VH1, VH2, VN1, and VN2 in module; AGND and DGND of sensor should be separated inside module and outside PCB single-point connection
- note 5** MCN/MCP AND MDN[0:3]/MDP[0:3] are five pairs of differential lines; lines should be as long as possible and equally spaced to be covered by the ground; differential line bottom (or TOP layer) should have a complete reference ground; PCB verification should tell PCB manufacturer which differential line requires manufacturer to carry out impedance control when making, impedance standard 100 ohm; error cannot exceed $\pm 10\%$.
- note 6** FPC using anti-jamming design

2.3 format and frame

The OS04J10 supports RAW RGB output with a 2/4-lane MIPI interface.

table 2-1 formats and frame rates

format	resolution	frame rate	methodology	10-bit output MIPI data rate	pixel clock
full resolution	2560x1440	60 fps	full resolution qualified pixel (2560+8)x(1440+8)	4-lane @ 720 Mbps/lane 2-lane @ 1440 Mbps/lane	72 MHz x2
1080p	1920x1080	79 fps	cropping qualified pixel (1920+8)x(1080+8)	4-lane @ 720 Mbps/lane 2-lane @ 1440 Mbps/lane	72 MHz x2
720p	1280x720	120 fps	2x2 binning qualified pixel (1280+8)x(720+8)	4-lane @ 720 Mbps/lane 2-lane @ 1440 Mbps/lane	72 MHz x2

2.4 MIPI interface

The OS04J10 supports a 2/4-lane MIPI transmitter interface and is capable of a data transfer rate of up to 1440 Mbps.

2.4.1 function related register configuration

table 2-2 MIPI function related registers (sheet 1 of 3)

address	register name	default value	R/W	description
P0:0x2C	MIPI_EN	0x00	RW	Bit[0]: mipi_en
P0:0x2D	R_CLK_TRAIL	0x07	RW	Bit[7:0]: r_clk_trail
P0:0x2E	R_EXIT	0x04	RW	Bit[7:0]: r_exit
P0:0x2F	R_CLK_PREPARE	0x04	RW	Bit[7:0]: r_clk_prepare
P0:0x30	H_SIZE_MIPI	0x0A	RW	Bit[3:0]: h_size_mipi[11:8]
P0:0x31	H_SIZE_MIPI	0x08	RW	Bit[7:0]: h_size_mipi[7:0]
P0:0x32	V_SIZE_MIPI	0x05	RW	Bit[2:0]: v_size_mipi[10:8]
P0:0x33	V_SIZE_MIPI	0xA8	RW	Bit[7:0]: v_size_mipi[7:0]
P0:0x34	HS_MODE	0x00	RW	Bit[1]: hs_mode_vf Bit[0]: hs_mode
P0:0x35	R_CLK_POST	0x0E	RW	Bit[7:0]: r_clk_post

table 2-2 MIPI function related registers (sheet 2 of 3)

address	register name	default value	R/W	description
P0:0x36	R_LPX_CK	0x07	RW	Bit[7:0]: r_lpx_ck
P0:0x37	R_HS_PREPARE	0x06	RW	Bit[7:0]: r_hs_prepare
P0:0x38	R_HS_ZERO	0x1A	RW	Bit[7:0]: r_hs_zero
P0:0x39	DATA_ID_REG	0x2B	RW	Bit[7:0]: data_id_reg
P0:0x3A	R_HS_TRAIL	0x08	RW	Bit[7:0]: r_hs_trail
P0:0x3B	R_CLK_ZERO	0x22	RW	Bit[7:0]: r_clk_zero
P0:0x3C	R_LPX_DAT	0x07	RW	Bit[7:0]: r_lpx_dat
P0:0x3D	TX_SPEED_AREA_SEL	0x03	RW	Bit[2:0]: tx_speed_area_sel
P0:0x3E	R_INIT	0x0B	RW	Bit[7:0]: r_init[15:8]
P0:0x3F	R_INIT	0x40	RW	Bit[7:0]: r_init[7:0]
P0:0x40	R_WAKEUP	0x00	RW	Bit[1:0]: r_wakeup[17:16]
P0:0x41	R_WAKEUP	0x86	RW	Bit[7:0]: r_wakeup[15:8]
P0:0x42	R_WAKEUP	0x88	RW	Bit[7:0]: r_wakeup[7:0]
P0:0x43	DC_TEST_LP_LK	0x3F	RW	Bit[7]: dc_test_hsd_c_en Bit[6]: dc_test_hsd_d_en Bit[5:4]: dc_test_lp_ck_en Bit[3:2]: dc_test_lp_d1_en
P0:0x44	DC_TEST_DATA_HS	0xFF	RW	Bit[7:0]: dc_test_data_hs
P0:0x45	FRAME_END_DLY	0x65	RW	Bit[7:0]: frame_end_dly
P0:0x46	MIPI_HSYNC_NUM	0x05	RW	Bit[7:0]: mipi_hsync_num
P0:0x47	MIPI_CTRL	0x20	RW	Bit[5]: mp_double Bit[4]: pause_ck Bit[3]: ls_mode Bit[2]: linit Bit[1]: ulp_mode Bit[0]: Testmode
P0:0x4A	HBLANK_READ	–	R	Bit[5:0]: blank_read[13:8]
P0:0x4B	HBLANK_READ	–	R	Bit[7:0]: blank_read[7:0]
P0:0x4C	MIPI_LS_CK	0x01	RW	Bit[3:2]: mipi_ls_start_num Bit[1:0]: mipi_ck_data
P0:0x4D	MIPI_CLK_INV	0x01	RW	Bit[0]: mipi_clk_inv
P0:0x52	CLK_TRAIL_MODE	0x01	RW	Bit[0]: clk_trail_mode

table 2-2 MIPI function related registers (sheet 3 of 3)

address	register name	default value	R/W	description
P0:0x53	MODE_10BIT	0x01	RW	Bit[0]: mode_10bit 0: 12-bit 1: 10-bit
P0:0x55	R_CLK_TIME	0x00	RW	Bit[6:0]: clk_time
P0:0x56	R_DATA_TIME	0x00	RW	Bit[5:0]: data_time

2.5 reset

The whole chip will be reset during power up. Manually applying a hardware reset (XSHUTDOWN = 0) upon power-up is recommended even though an on-chip power on reset is included. The hardware reset is active low with an asynchronous design. The reset pulse width should be greater than or equal to 2 ms.

2.5.1 power on reset generation

The power on reset can be controlled from an external pin. Additionally, in this sensor, a power on reset is generated after the core power becomes stable.

2.5.2 software reset

A reset can also be initiated via the SCCB interface by writing:

0xFC = 0x01

Then, all registers will be reset to their default values.

2.6 standby mode

There are two suspend modes available for the OS04J10:

- **hardware standby**
- **software standby**

2.6.1 hardware standby

To initiate hardware standby mode, XSHUTDOWN must be tied to low. When this occurs, the OS04J10 internal device clock is halted even when the external clock source is still clocking and all internal counters are reset.

2.6.2 software standby

Executing a software power down (register P0:0x09 = 0x07) through the SCCB interface suspends internal circuit activity, but does not halt the device clock. All register content is maintained in standby mode. During the resume state, all registers are restored to their original values.

table 2-3 hardware and standby descriptions

mode	description
hardware standby with XSHUTDOWN	1. Enabled by pulling XSHUTDOWN low. 2. Power down all blocks. 3. Register values are reset to default values. 4. No SCCB communication. 5. Minimum power consumption.
software standby	1. Default mode after power on reset. 2. Power down all blocks except SCCB. 3. Register values are maintained. 4. SCCB communication is available. 5. Low power consumption. 6. GPIO can be configured as high/low/tri-state.

2.7 system clock control

2.7.1 MPLL/DPLL clock scheme

The OS04J10 has a two PLL structures: one for MIPI to minimize EMI and the other for ADC. Registers for clock configuration are shown in **table 2-4**.

MIPI PLL (MPLL) generates a default 72 MHz pixel clock and a 1440 MHz MIPI serial clock from a 6~27 MHz input clock. The VCO range is 500~1500 MHz. A programmable clock divider is provided to generate different frame rate timings. The internal function blocks of MPLL are shown in **figure 2-3**.

DPLL generates a default 756 MHz counter clock, from a 6~27 MHz input clock, for internal ADC operation. The VCO range is 250~1500 MHz. A programmable clock divider is provided to generate different frame rate timings. The internal function blocks of DPLL are shown in **figure 2-4**.

2.7.2 clock configuration

table 2-4 clock configuration related registers (sheet 1 of 3)

address	register name	default value	R/W	description
P0:0x11	P0_COMC06	0x00	RW	Bit[7:0]: mpll_dsm[7:0]
P0:0x12	P0_COMC07	0x42	RW	Bit[7:4]: mpll_dsm[19:16] Bit[3]: dpll_cp_clk_disab_sel Bit[2]: mpll_rst_sync_en Bit[1]: mpll_sscg_cntstep_x3 Bit[0]: mpll_sscg_en

table 2-4 clock configuration related registers (sheet 2 of 3)

address	register name	default value	R/W	description
P0:0x13	P0_COMC08	0x06	RW	Bit[7]: mppll_bp_vco_sync Bit[6]: mppll_dither_en Bit[5]: mppll_div_pwd Bit[4]: mppll_frac_en Bit[3]: mppll_loop_pwd Bit[2:0]: mppll_sscg_cntck
P0:0x14	P0_COMC09	0x1E	RW	Bit[7]: vdd_clamp_dn_en Bit[6]: vdd_clamp_up_en Bit[5:4]: mppll_cp_clk_post_sel Bit[3:2]: mppll_cp_clk_pre_sel Bit[1]: mppll_cp_clk_disab_sel Bit[0]: mppll_pclk_disab_sel
P0:0x15	P0_COMC0A	0x01	RW	Bit[7:6]: dpll_cp_sel Bit[5]: dpll_div_pwd Bit[4]: dpll_band_sel Bit[3]: dpll_en_exti_sel Bit[2]: dpll_predivp_sel Bit[1]: dpll_rst_sync_sel Bit[0]: dpll_vcoby2_sel
P0:0x16	DPLL_NC_SEL	0x00	RW	Bit[1:0]: dpll_nc_sel[9:8]
P0:0x17	DPLL_NC_SEL	0x7E	RW	Bit[7:0]: dpll_nc_sel[7:0]
P0:0x18	DPLL_MC_SEL	0x05	RW	Bit[2:0]: dpll_mc_sel
P0:0x19	P0_COMC0B	0x60	RW	Bit[7]: dpll_row_clk_disab_sel Bit[6:4]: dpll_row_clk_post_sel Bit[3:2]: dpll_cnt_clk_sel Bit[0]: dpll_byp_sel
P0:0x1A	P0_COMC0C	0x65	RW	Bit[7]: dpll_cp_clk_pre_sel Bit[6]: dpll_pclk_disab_sel Bit[5:4]: dpll_pclk_post_sel Bit[3:2]: dpll_pclk_pre_sel Bit[1:0]: dpll_row_clk_pre_sel
P0:0x1B	P0_COMC12	0x21	RW	Bit[7:4]: mppll_cp_sel Bit[3:0]: mppll_sscg_cntstep
P0:0x1C	MPLL_NC_SEL	0x00	RW	Bit[1:0]: mppll_nc_sel[9:8]
P0:0x1D	MPLL_NC_SEL	0xF0	RW	Bit[7:0]: mppll_nc_sel[7:0]
P0:0x1E	MPLL_MC_SEL	0x05	RW	Bit[2:0]: mppll_mc_sel
P0:0x1F	MPLL_COMC02	0x21	RW	Bit[5:4]: mppll_pclk_post_sel Bit[3]: mppll_bclk_sel Bit[2:0]: mppll_pclk_pre_sel

table 2-4 clock configuration related registers (sheet 3 of 3)

address	register name	default value	R/W	description
P0:0x20	MPLL_COMC03	0x01	RW	Bit[5]: mpll_byp_sel Bit[4]: mpll_predivp_sel Bit[2:0]: mpll_phy_clk_sel
P0:0x21	P0_COMC15	0x27	RW	Bit[6:4]: mpll_opa_sel Bit[2:0]: dpll_cp_clk_post_sel
P0:0x24	P0_COMC0D	0x40	RW	Bit[7]: mipi_valid_sel Bit[6]: mipi_vb_psv_byp Bit[5:4]: mipi_vof_sel[1:0] Bit[3]: mipi_oterm_en Bit[2:0]: mipi_vcm_sel
P0:0x25	P0_COMC0E	0x01	RW	Bit[7:6]: mipi_ck_skew_sel Bit[5]: mipi_eq_en Bit[4]: mipi_hsen_skew_sel Bit[3]: mipi_lp_rtime_dis Bit[2]: mipi_lp_sr_sel Bit[1:0]: mipi_lptx_sel
P0:0x26	P0_COMC0F	0x00	RW	Bit[7:6]: mipi_d0_skew_sel Bit[5:4]: mipi_d1_skew_sel Bit[3:2]: mipi_d2_skew_sel Bit[1:0]: mipi_d3_skew_sel
P0:0x27	CLK_MODE	0x07	RW	Bit[2]: dclk_otp_dpc_gating_en Bit[1]: dclk_chn_gating_en Bit[0]: dclk2x_gating_en
P0:0x28	CLK_MODE	0xFE	RW	Bit[6]: bclk2x_gating_en Bit[5]: dclkin_gating_en Bit[4]: dclk_dpc_gating_en Bit[3]: dclk_blc_gating_en Bit[2]: dclk_mipi_gating_en Bit[1]: bclk_gating_en Bit[0]: pclk_in_sel 0: pclk_in 1: EXTCLK
P0:0x29	CLK_MODE	0x20	RW	Bit[5:4]: row_clk_sel Bit[2]: dclk_ctrl Bit[1:0]: timer_clk_ctrl

2.7.3 PLL clock scheme

The internal function blocks of the PLLs are shown in **figure 2-3** and **figure 2-4**.

figure 2-3 MPLL internal function block diagram

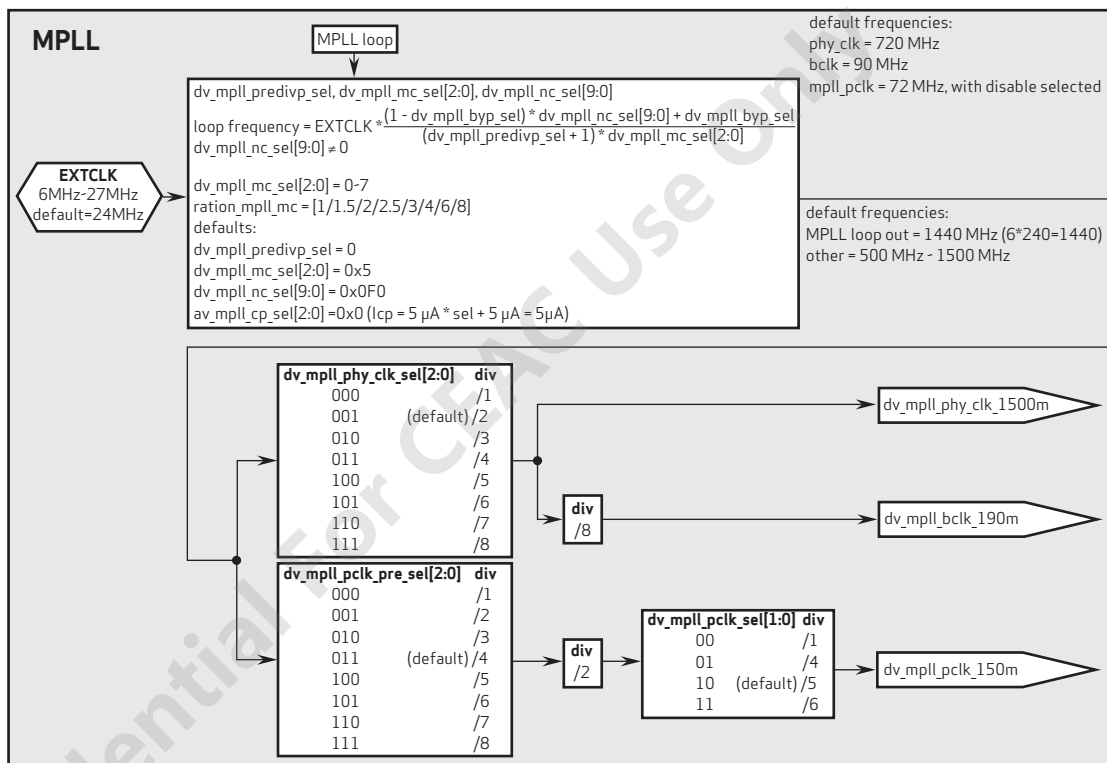
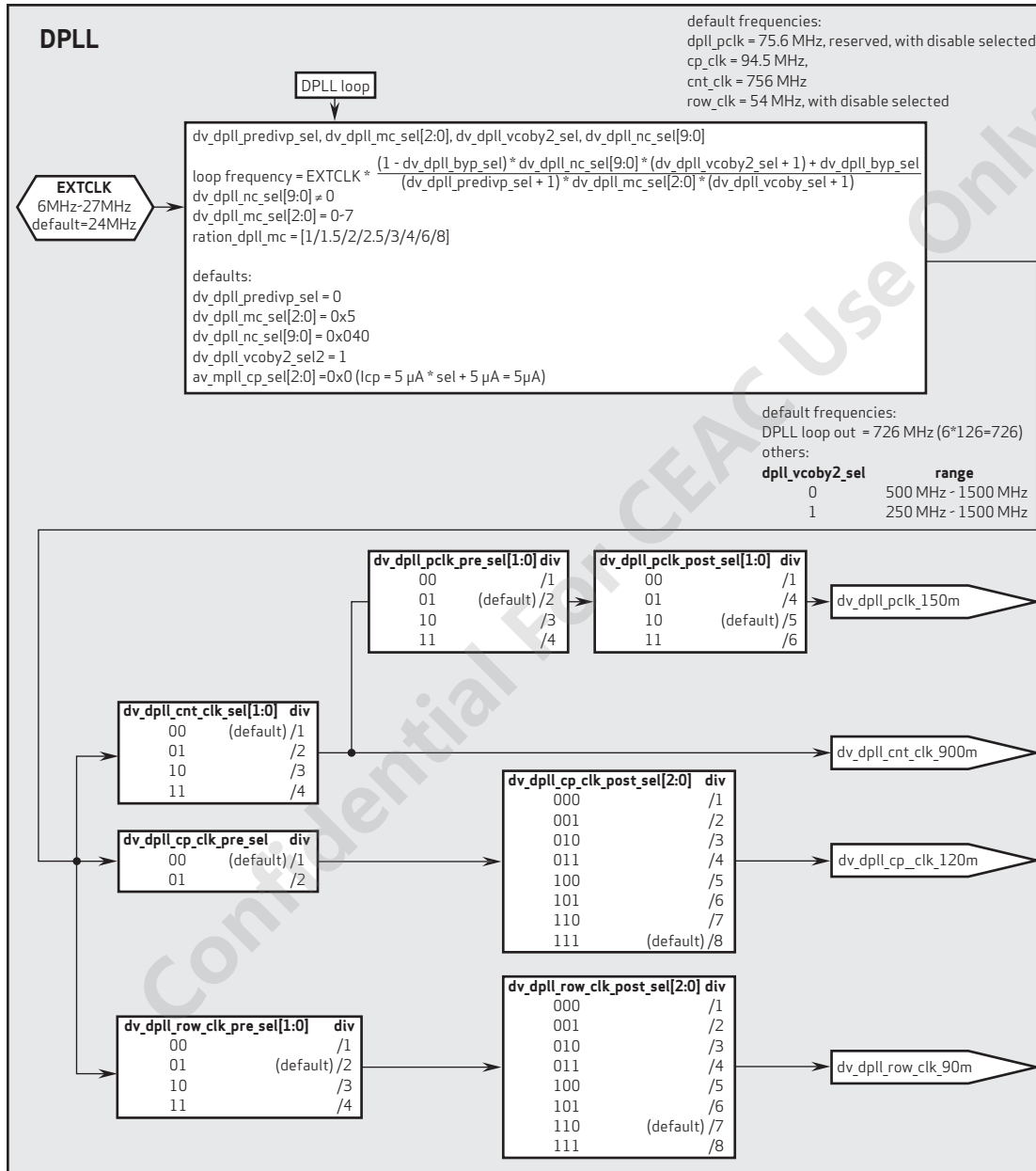


figure 2-4 DPLL internal function block diagram



2.8 power management

In power down mode, all internal clocks are stopped and all sensor modules are powered off. The register values will all be reset in power down mode.

In soft reset mode, all register values will be reset, but all internal clocks are kept and all sensor modules work normally.

In low power wake mode, all internal clocks are stopped and all sensor modules are powered off. The register values will all be kept in low power wake mode.

2.9 data lane parameters

The D-PHY converts parallel data from CSI_TOP to serial data that is compatible with MIPI protocol. Detailed time sequences are shown in **figure 2-5** and **figure 2-6**.

figure 2-5 high-speed data transmission bursts

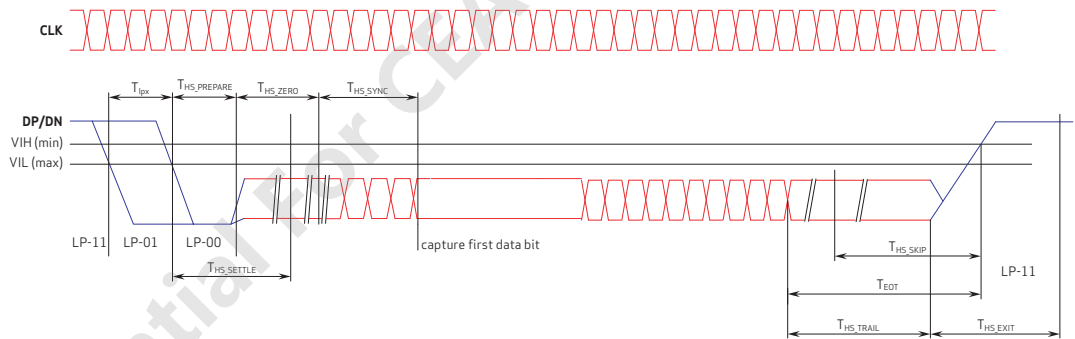


figure 2-6 switching the clock lane between clock transmission and low-power mode

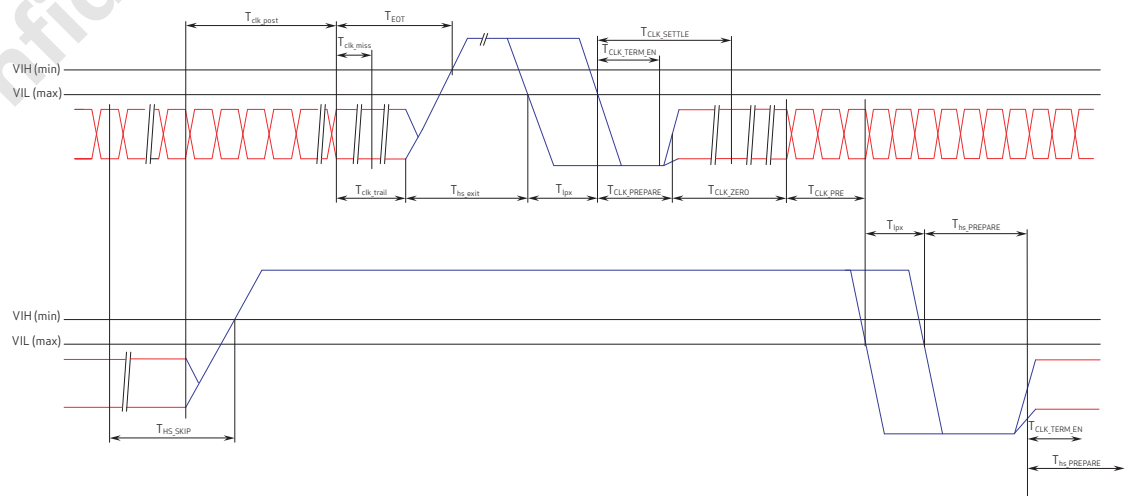


table 2-5 data lane parameters

parameter	target	min	max
$T_{\text{HS-PREPARE}}$	time that transmitter drives data lane LP-00 line state immediately before HS-0 line state starts HS transmission	$40 \text{ ns} + 4 \cdot \text{UI}$	$85 \text{ ns} + 6 \cdot \text{UI}$
$T_{\text{HS-PREPARE}} + T_{\text{HS-ZERO}}$	$T_{\text{HS-PREPARE}}$ + time that transmitter drives HS-0 state prior to transmitting sync sequence	$145 \text{ ns} + 10 \cdot \text{UI}$	
$T_{\text{HS-SETTLE}}$	time interval which HS receive shall ignore any data lane HS transitions, starting from beginning of $T_{\text{HS-PREPARE}}$	$85 \text{ ns} + 6 \cdot \text{UI}$	$145 \text{ ns} + 10 \cdot \text{UI}$
$T_{\text{HS-SKIP}}$	time interval which HS-RX should ignore any transitions on data lane, following an HS burst end point of interval is defined as beginning of LP-11 state following HS burst	40 ns	$55 \text{ ns} + 4 \cdot \text{UI}$
$T_{\text{HS-TRAIL}}$	time that transmitter drives HS-0 state after last payload clock bit of an HS transmission burst	max ($n \cdot 8 \cdot \text{UI}$, $60 \text{ ns} + n \cdot 4 \cdot \text{UI}$)	
T_{LPX}	transmitted length of any low-power state period	50 ns	
T_{WAKEUP}	time that a transmitter drives a Mark-1 state prior to a stop state in order to initiate an exit from ULPS	1 ms	

2.10 clock lane parameters

table 2-6 clock lane parameters

parameter	target	min	max
$T_{\text{CLK-MISS}}$	timeout for receiver to detect absence of clock transmission and disable clock lane HS-RX	60 ns	
$T_{\text{CLK-POST}}$	time that transmitter continues to send HS clock after last associated data lane has transitioned to LP mode interval is defined as period from end of $T_{\text{HS-TRAIL}}$ to beginning $T_{\text{CLK-TRAIL}}$	$60 \text{ ns} + 52 \cdot \text{UI}$	
$T_{\text{CLK-PRE}}$	time that HS clock will be driven by transmitter prior to any associated data lane beginning transition from LP to HS mode	$8 \cdot \text{UI}$	
$T_{\text{CLK-PREPARE}}$	time that transmitter drives clock lane LP-00 line state immediately before HS-0 line state starts HS transmission	38 ns	95 ns
$T_{\text{CLK-SETTLE}}$	time interval which HS receive will ignore any clock lane HS transmissions, starting from beginning of $T_{\text{CLK-PREPARE}}$	95 ns	300 ns
$T_{\text{CLK-TRAIL}}$	time that transmitter drives HS-0 state after last payload clock bit of an HS transmission burst	60 ns	
$T_{\text{CLK-PREPARE}} + T_{\text{CLK-ZERO}}$	$T_{\text{CLK-PREPARE}}$ + time that transmitter drives HS-0 state prior to starting clock	300 ns	
T_{EOT}	transmitted time interval from start of $T_{\text{CLK-TRAIL}}$ or $T_{\text{HS-TRAIL}}$, to start of LP-11 state following an HS burst		$105 \text{ ns} + n \cdot 12 \cdot \text{UI}$

2.11 power up/off sequence

2.11.1 power up sequence

figure 2-7 power up sequence timing diagram

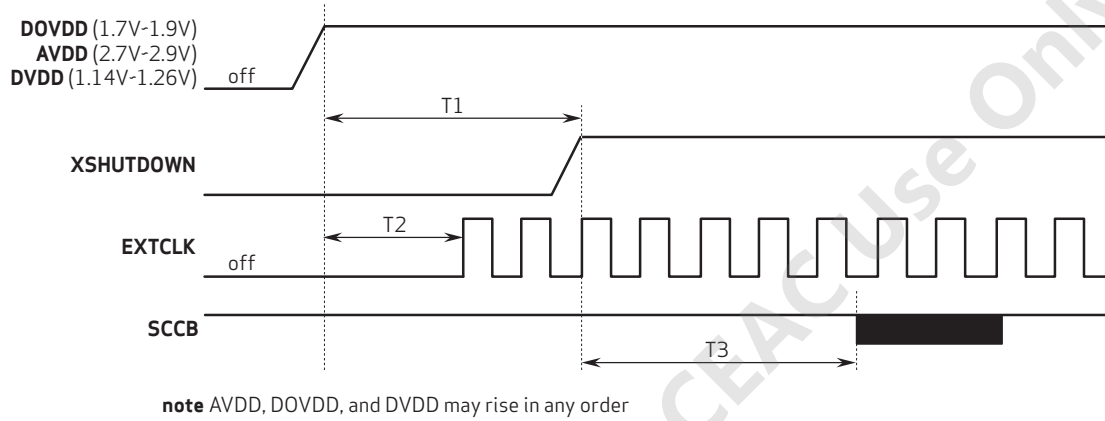


table 2-7 power up sequence timing specifications

symbol	description	min	unit
T1	delay from power supply on to XSHUTDOWN pull up	5	ms
T2	delay from power supply on to EXTCLK	0	ms
T3	delay from XSHUTDOWN pull up to SCCB initialization	8	ms

2.11.2 power off sequence

figure 2-8 power off sequence timing diagram

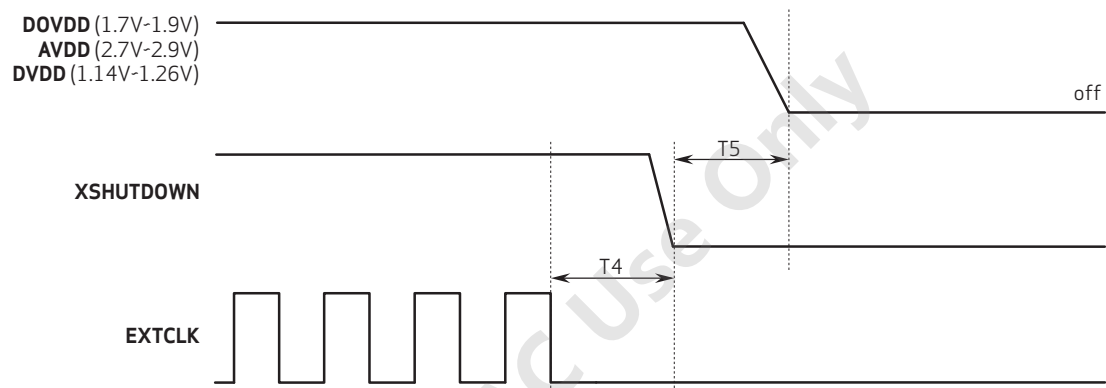


table 2-8 power off sequence timing specifications

symbol	description	min	unit
T4	delay from EXTCLK off to sensor power down	0	ms
T5	delay from sensor power down to power supply off	0	ms

2.12 SCCB interface

2.12.1 single read and single write

The OS04J10 SCCB write address and read address can be selected by the SID pin. When the pin is set high, the write address is 0x7A and the read address is 0x7B. When the pin is set low, the write address is 0x78 and the read address is 0x79.

A typical read or write sequence begins by the master sending a start bit. After the start bit, the master sends the slave device's 8-bit address. The last bit of the address determines if the request will be a read or a write, where a '0' indicates a write and a '1' indicates a read. The slave device acknowledges its address by sending an acknowledge bit back to the master.

If the request was a write, the master then transfers the 8-bit register address to which a write should take place. The slave sends an acknowledge bit to indicate that the register address has been received. The master then transfers the data 8 bits at a time, with the slave sending an acknowledge bit after each 8 bits. The master stops writing by sending a start or stop bit.

A typical read sequence is executed as follows. First, the master sends the write-mode slave address and 8-bit register address just as in the write request. The master then sends a start bit and the read-mode slave address. The master then clocks out the register data 8 bits at a time. The master sends an acknowledge bit after each 8-bit transfer. The data transfer is stopped when the master sends a no-acknowledge bit.

figure 2-9 illustrates the OS04J10 single read sequence and single write sequence.

figure 2-9 SCCB read and write message description (SID pin set high)

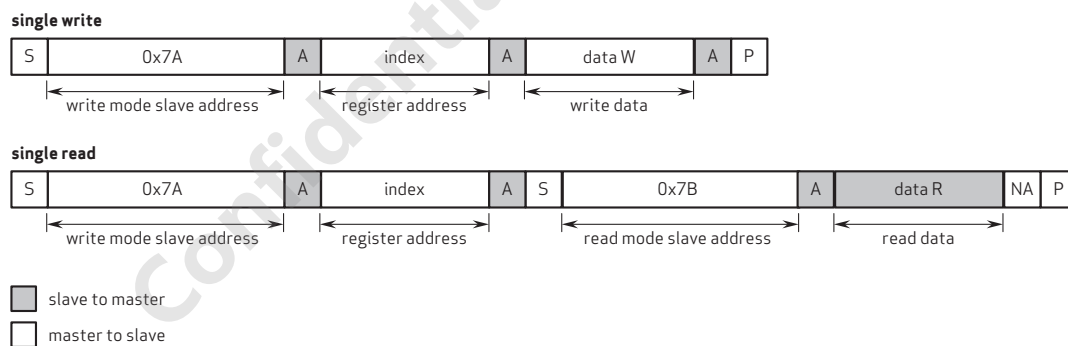
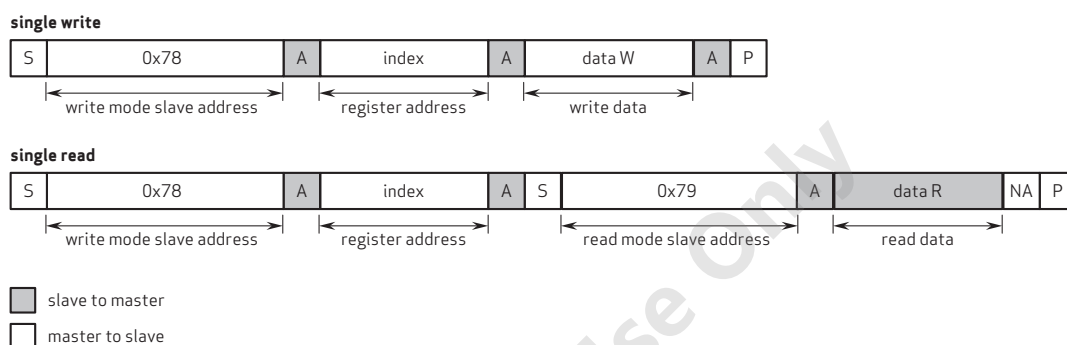


figure 2-10 SCCB read and write message description (SID pin set low)



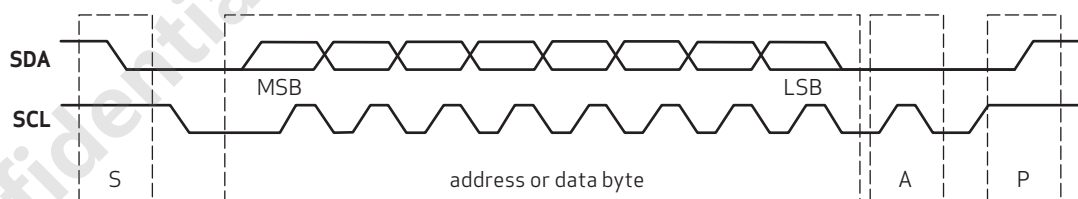
2.12.2 data bit transfer

One data bit is transferred during each clock pulse. The serial clock pulse is provided by the master. The data must be stable during the high period of the serial clock. It can only change when the serial clock is low. Data is transferred 8 bits at a time, followed by an acknowledge bit.

2.12.3 acknowledge bit

The OS04J10 will hold the value of the SDA pin to logic '0' during the logic '1' state of the acknowledge clock pulse on SCL.

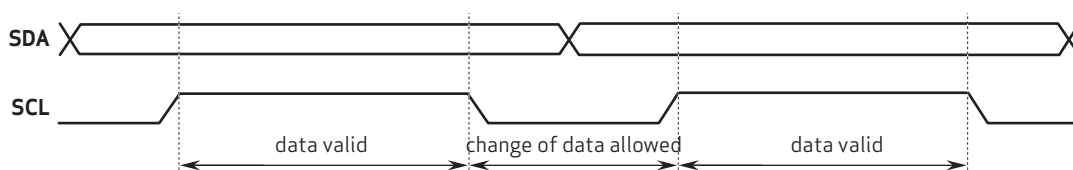
figure 2-11 SCCB acknowledge bit diagram



2.12.4 data valid

The master must ensure that the data is stable during the logic 1 state of the SCL pin. All transitions on the SDA pin can only occur when the logic level on the SCL pin is '0'.

figure 2-12 SCCB data transport diagram



2.12.5 timing parameter

figure 2-13 SCCB interface timing diagram

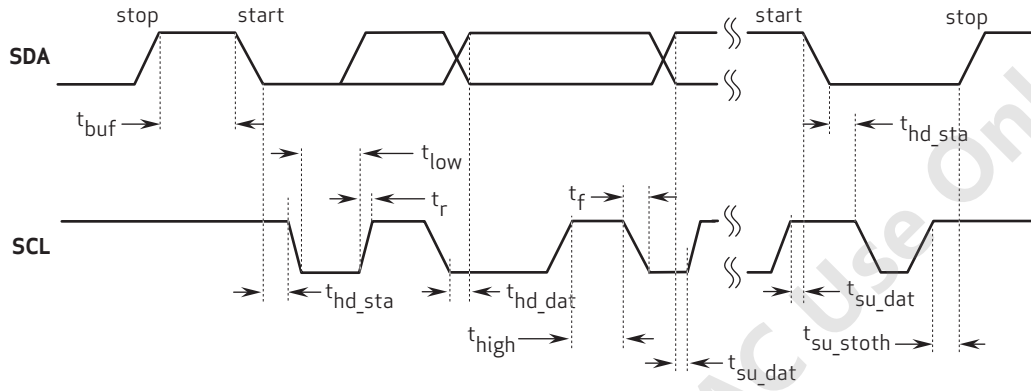


table 2-9 SCCB interface timing specifications

symbol	parameter	min	max	unit
f_{scl}	SCL clock frequency	—	400	kHz
t_{buf}	bus free time between a stop and a start	1.3	—	μs
t_{hd_sta}	hold time for a repeated start	0.6	—	μs
t_{low}	low period of SCL	1.3	—	μs
t_{high}	high period of SCL	0.6	—	μs
t_{su_sta}	setup time for a repeated start	0.6	—	μs
t_{hd_dat}	data hold time	0	—	μs
t_{su_dat}	data setup time	0.1	—	μs
t_r	rise time of SCL, SDA	—	0.3	μs
t_f	fall time of SCL, SDA	—	0.1	μs
t_{su_sto}	setup time for a stop	0.6	—	μs
t_{aa}	SCL low to data out valid	0.3	—	μs
t_{dh}	data out hold time	0.2	—	μs
C_b	capacitive load of bus line (SCL, SDA)	—	—	pf

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3 block level description

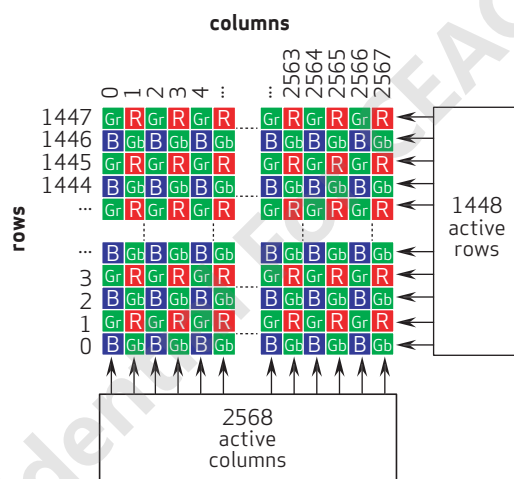
3.1 pixel array structure

The OS04J10 sensor has an image array of 2568 columns by 1448 rows (3,718,464 pixels). **figure 3-1** shows a cross-section of the image sensor array.

The color filters are arranged in a Bayer pattern. The primary color GR/BG array is arranged in line-alternating fashion. Of the 3,718,464 pixels, all 3,718,464 pixels (2568x1448) are active pixels and can be output.

The sensor array design is based on a field integration readout system with line-by-line transfer and an electronic shutter with a synchronous pixel readout scheme.

figure 3-1 sensor array region color filter layout



3.2 binning

The 2x binning function that is embedded in the OS04J10 has an image output mode conversion function. Both horizontal binning and vertical binning are averaged.

The basic principle of the binning function is interlaced data collection, which means that the first line data is collected, then the adjacent second line is ignored, and the third line is collected, and so on. For example, in the blue pixel 2x2 binning, the number 0 and number 2 blue pixels of the first line and the number 0 and number 2 blue pixels of the third line can be considered as four factors of the new blue pixel.

By doing so, the line-mismatch phenomenon and saw tooth phenomenon can be avoided. Also, the signal-to-noise ratio (SNR) of the pixel can be increased when the binning function is on, and the chip asks for a lower power supply.

figure 3-2 example of 2x2 binning

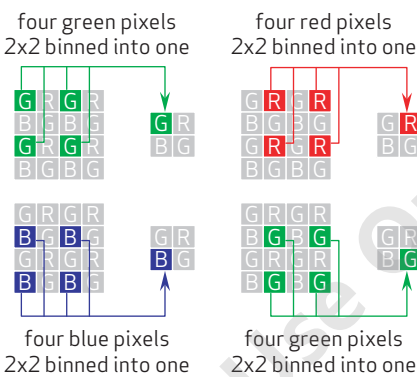


table 3-1 binning-related registers

address	register name	default value	R/W	description	
P1:0x32	VBINNING_EN	0x00	RW	Bit[1]: Bit[0]:	Enable mono binning Enable color binning Must be configured at same time as register P3:0x0E for color binning
P3:0x0E	BINNING_EN	0x00	RW	Bit[1]: Bit[0]:	Enable column binning average Enable binning Must be configured at same time as register P1:0x32 for color binning

3.3 analog amplifier

When the column sample/hold circuit has sampled one row of pixels, the pixel data will shift out one-by-one into an analog amplifier.

3.4 12-bit A/D converters

The balanced signal is then digitized by the on-chip 12-bit ADC.

3.5 high dynamic range (HDR) mode

HDR mode increases image dynamic range by capturing two exposures of a similar scene and then combining them into one single image. The OS04J10 uses staggered HDR.

In staggered HDR mode, long/short exposure frames are overlapping with each other. This reduces the timing delay between different exposure frames, which will combine into one HDR frame. It also reduces the frame/line buffer needed for the backend chip.

In HDR mode, long and short frames are transmitted through virtual channels, as shown in **figure 3-3**. In HDR dummy mode, long and short frames are transmitted as one frame through the same virtual channel, as shown in **figure 3-4**.

figure 3-3 long and short frame transmission in HDR mode

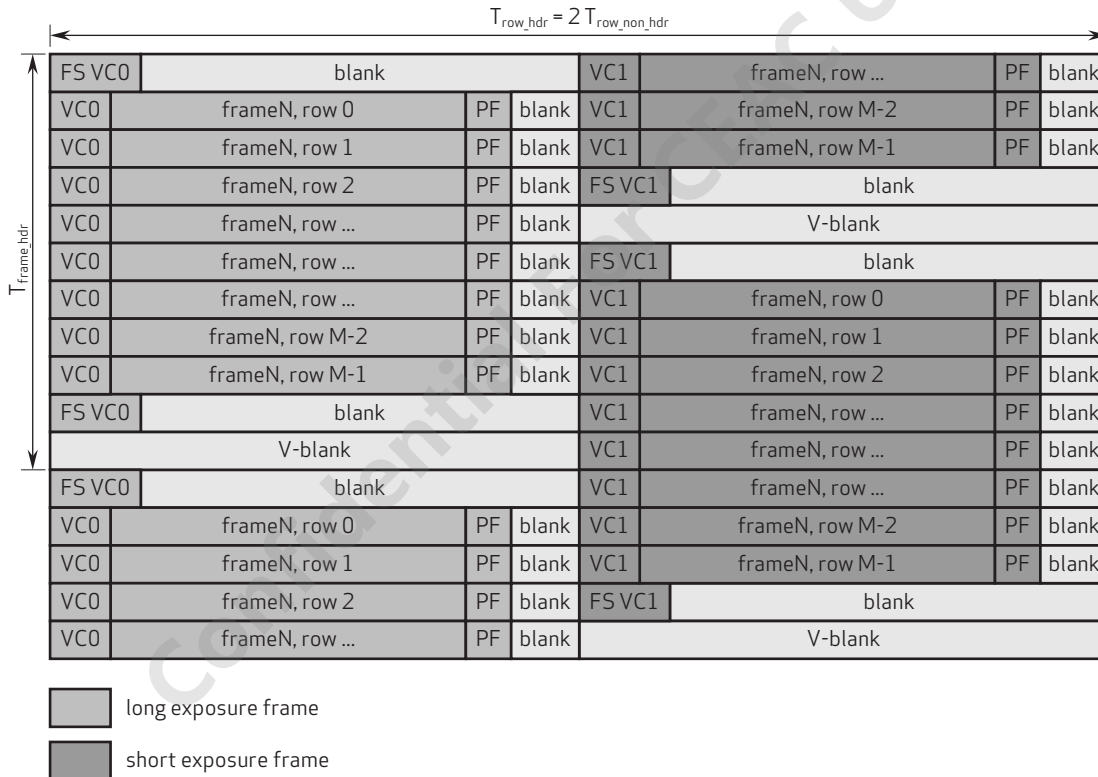
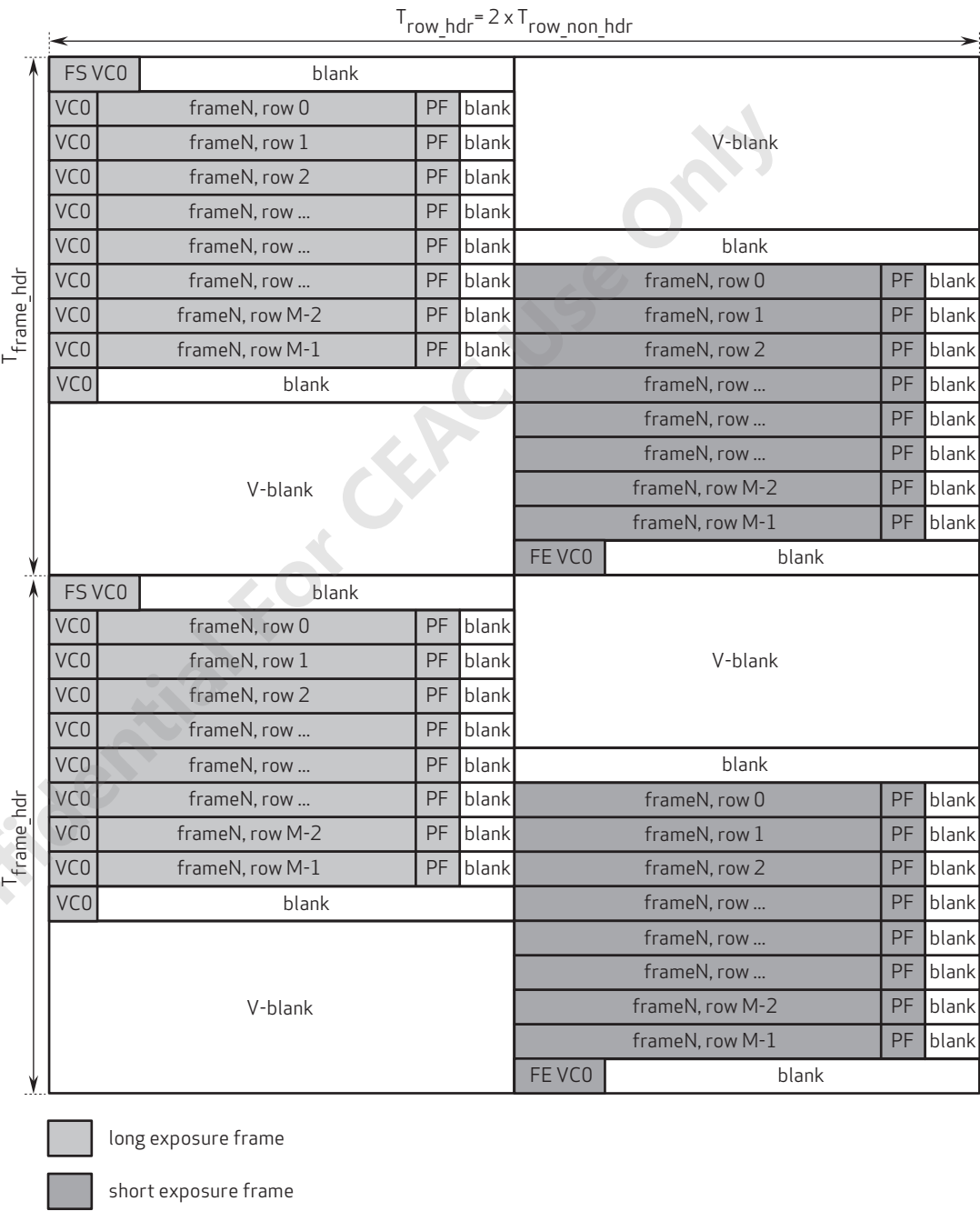


figure 3-4 long and short frame transmission without virtual channels in HDR dummy mode



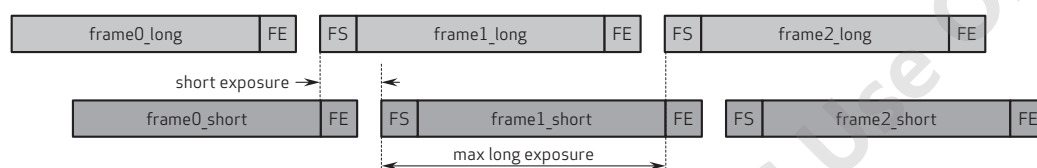
The OS04J10 supports two HDR modes:

- variable maximum short exposure HDR mode
- fixed maximum short exposure HDR mode

3.5.1 variable maximum short exposure HDR mode

The timing delay between long frame and short frame is short exposure.

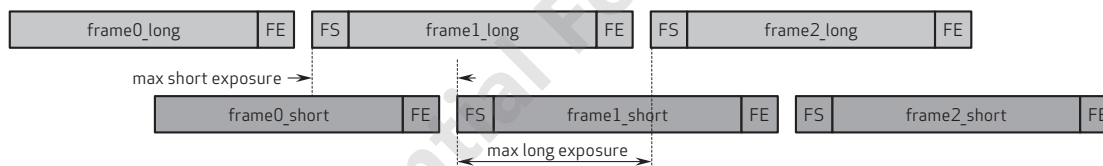
figure 3-5 variable maximum short exposure HDR mode



3.5.2 fixed maximum short exposure HDR mode

The timing delay between long frame and short frame is maximum short exposure and the maximum short exposure is set by registers.

figure 3-6 fixed maximum short exposure HDR mode



3.6 dual conversion gain

The OS04J10 supports dual conversion gain (DCG). DCG is a pixel level gain method that divides the gain into two levels, high conversion gain (HCG) and low conversion gain (LCG). The HCG and LCG frames are transmitted through different virtual channels, similar to the long and short HDR frames. The OS04J10 also supports DCG combine mode, which composes HCG frame and LCG frame into one frame, output at 12/16-bit.

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4 image sensor core digital functions

4.1 mirror and flip

The OS04J10 provides mirror and flip readout modes, which respectively reverse the sensor data readout order horizontally and vertically (see **figure 4-1**).

figure 4-1 mirror and flip samples

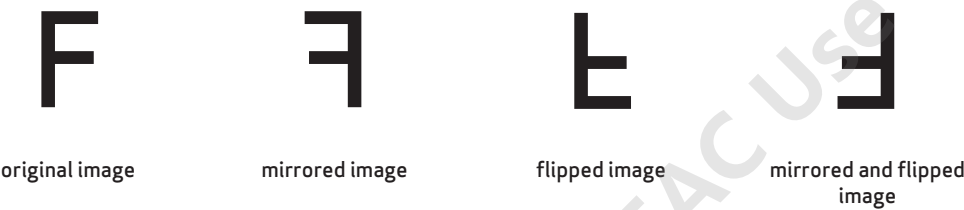


table 4-1 mirror and flip registers

address	register name	default value	R/W	description	
P1:0x12	FLIP/MIRROR	0x00	RW	Bit[3]:	Preserve Bayer pattern on mirror
				Bit[2]:	Preserve Bayer pattern on flip
				Bit[1:0]:	Image mirror/flip
				00:	Normal
				01:	Vertical flip
				10:	Horizontal mirror
				11:	Mirror and flip

4.2 windowing

The embedded windowing function extracts an image windowing area by defining four parameters, including horizontal start, horizontal width, vertical start, and vertical height. By properly setting the parameters, the portions within the sensor array size can be cropped as a visible area. The windowing function will not conflict with the mirror and flip function as shown in **figure 4-2**. Horizontal and vertical size is automatically divided when binning, configure them by full size.

figure 4-2 windowing diagram

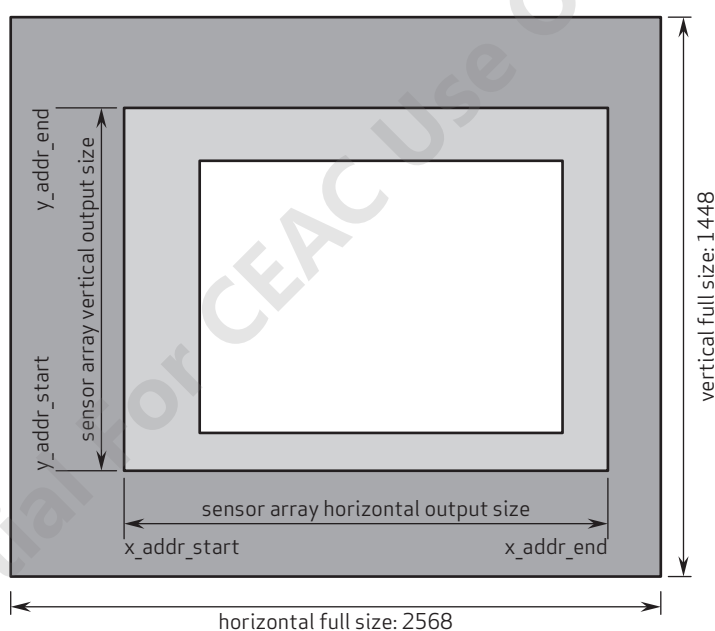


table 4-2 digital windowing control registers (sheet 1 of 2)

address	register name	default value	R/W	description
P3:0x00	HSTART_CIS	0x00	RW	Bit[3:0]: hstart_cis[11:8]
P3:0x01	HSTART_CIS	0x00	RW	Bit[7:0]: hstart_cis[7:0]
P3:0x02	HSIZE_CIS	0x0A	RW	Bit[3:0]: hsize_cis[11:8]
P3:0x03	HSIZE_CIS	0x08	RW	Bit[7:0]: hsize_cis[7:0]
P3:0x04	VSTART_CIS	0x00	RW	Bit[2:0]: vstart_cis[10:8]
P3:0x05	VSTART_CIS	0x20	RW	Bit[7:0]: vstart_cis[7:0]
P3:0x06	VSIZ_CIS	0x05	RW	Bit[2:0]: vsize_cis[10:8]
P3:0x07	VSIZ_CIS	0xA8	RW	Bit[7:0]: vsize_cis[7:0]

table 4-2 digital windowing control registers (sheet 2 of 2)

address	register name	default value	R/W	description
P3:0x16	HSTART_WIN	0x00	RW	Bit[3:0]: hstart_win[11:8]
P3:0x17	HSTART_WIN	0x00	RW	Bit[7:0]: hstart_win[7:0]
P3:0x18	HSIZE_WIN	0x0A	RW	Bit[3:0]: hsize_win[11:8]
P3:0x19	HSIZE_WIN	0x08	RW	Bit[7:0]: hsize_win[7:0]
P3:0x1A	VSTART_WIN	0x00	RW	Bit[2:0]: vstart_win[10:8]
P3:0x1B	VSTART_WIN	0x00	RW	Bit[7:0]: vstart_win[7:0]
P3:0x1C	VSIZE_WIN	0x05	RW	Bit[2:0]: vsize_win[10:8]
P3:0x1D	VSIZE_WIN	0xA8	RW	Bit[7:0]: vsize_win[7:0]

table 4-3 analog windowing control registers

address	register name	default value	R/W	description
P1:0x04	VSTART	0x00	RW	Bit[2:0]: Analog window vstart[10:8]
P1:0x05	VSTART	0x20	RW	Bit[7:0]: Analog window vstart[7:0]
P1:0x08	VSIZE	0x05	RW	Bit[2:0]: Analog window vsize[10:8]
P1:0x09	VSIZE	0xA8	RW	Bit[7:0]: Analog window vsize[7:0]

4.3 test pattern

A color bar is available for testing purposes.

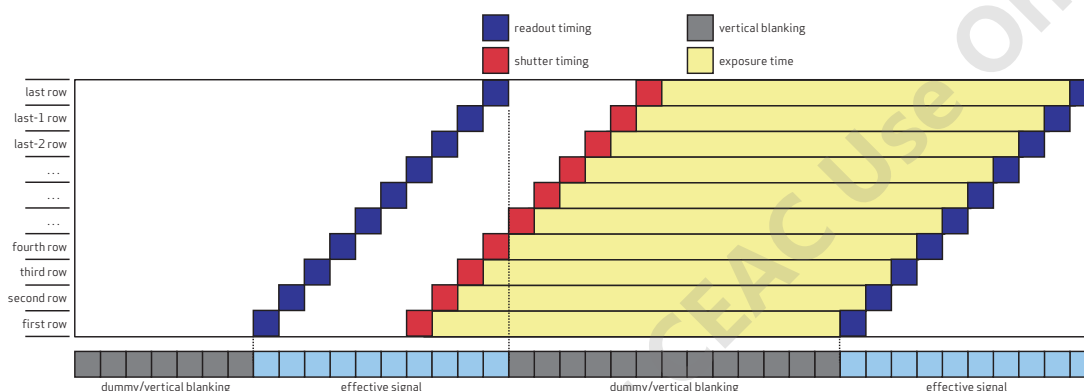
figure 4-3 color bar



4.4 shutter and integration time settings and continuous exposure control

The OS04J10 has a variable electronic shutter function that can control the integration time in row, or half-row, units. In addition, the OS04J10 performs rolling shutter operation in which electronic shutter and readout are performed sequentially for each row.

figure 4-4 shutter and integration time settings and continuous exposure control



The integration time can be controlled by varying the electronic shutter time. In the electronic shutter setting, integration time is controlled by the EXP register ({P1:0x0E, P1:0x0F}).

4.4.1 exposure/gain control

The OS04J10 long frame exposure registers are {P1:0x0E, P1:0x0F}, the short frame exposure registers are {P1:0x1A, P1:0x1B}. The unit is Tline. Minimum exposure is 2 Tline. Maximum exposure is VTS-8. If exposure is longer than VTS-8, the OS04J10 will automatically prolong the current VTS. After registers P1:0x0E and P1:0x0F are set, register P1:0xFE (RESTART) must be set to 0x02 to trigger the new exposure.

The OS04J10 supports automatic gain control (AGC). The long frame AGC registers are {P1:0x24} and the short frame AGC registers are {P1:0xE7}. The step size of AGC gain is 1/16: 0x010 is 1x AGC gain, and the maximum AGC gain is 0xFF = 15.5x. Changes to AGC gain must be triggered by setting register P1:0xFE to 0x02.

The OS04J10 supports digital gain. The long frame digital gain registers are {P1:0x1F, P1:0x20}, and the short frame digital gain registers are {P1:0xE4, P1:0xE5}. The step size of digital gain is 1/32: 0x040 is 1x digital gain, and the maximum digital gain is 0x7FE = 32x. Changes to digital gain must be triggered by setting register P1:0xFE to 0x02.

Below is an example setting for exposure/AGC gain/digital gain control:

```
78 FD 01
78 0E 01
78 0F 54 ; exposure time 0x0154
78 24 80 ; AGC gain 8x
78 20 80 ; digital gain 2x
78 FE 02 ; trigger the new exposure/gain
```

The exposure/AGC gain/digital gain will be valid for an N+2 frame when P1:0xFE is set to 0x02.

4.4.2 HTS/VTs

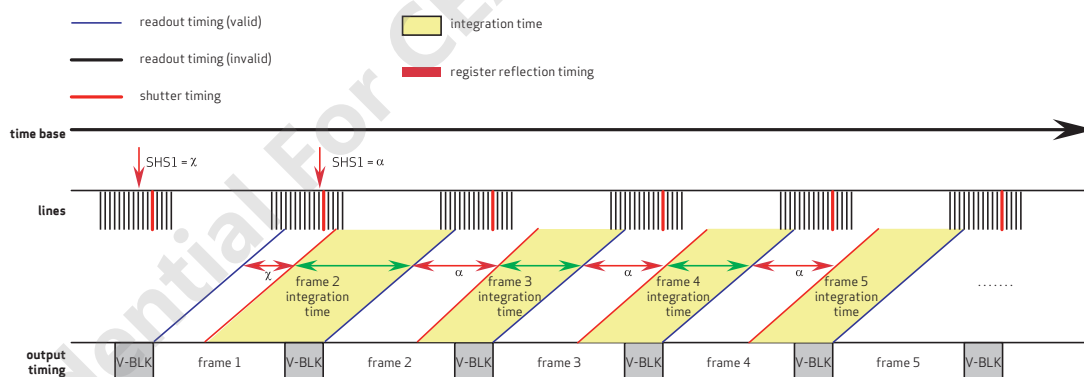
The HTS/VTs registers are read only registers. Registers P1:0x2A and P1:0x2B are HTS registers. The unit is row_clk. Registers P1:0x2F and P1:0x30 are VTS registers. The unit is Tline. Registers P1:0x14 and P1:0x15 are Vblank registers. To modify VTS, insert dummy lines with registers P1:0x14 and P1:0x15. Changes to Vblank must be triggered by setting register P1:0xFE to 0x02.

For example, if the current dummy line is 4 lines (P1:0x14 = 0x00 and P1:0x15 = 0x04) and the user wants to insert 16 dummy lines, the following setting is recommended:

```
78 FD 01
78 14 00
78 15 14 ;    dummy lines set to 20 (4 + 16)
78 FE 02 ;    trigger new Vblank
```

It is not recommended to change the HTS. If a change to the HTS is desired, please contact the local OmniVision FAE for support.

figure 4-5 integration time control within a frame



4.5 FSIN

Data output of the OS04J10 can be synchronized via FSIN pad. The OS04J10 can be either a master or slave device in FSIN mode. After synchronization, the time difference between the data output of master and slave is less than 1 row time.

4.5.1 FSIN common control

table 4-4 FSIN control registers (sheet 1 of 2)

address	register name	default value	R/W	description
P1:0xD3	EVSYNC_CTRL	0x00	RW	Bit[7]: exter_sync_inv Bit[5]: rst_colrow_en Bit[4]: slave_preshutter_en Bit[2]: sync_no_wait_en Bit[1]: exter_slave_en Bit[0]: exter_master_en
P1:0xD5	FSIN_OUT_FORMAT	0x00	RW	Bit[0]: fsin_out_format 0: fsin_out 1: vsync_isp
P1:0xD6	EXTER_SYNC_FRAME_NUM	0x01	RW	Bit[7:0]: exter_sync_frame_num
P1:0xD7	SLAVE_SHUTTER_PERIOD	0x00	RW	Bit[7:0]: Slave shutter period[7:0] Slave preshutter reset row address
P1:0xD8	SLAVE_SHUTTER_PERIOD	0x01	RW	Bit[3:0]: Slave shutter period[11:8] Slave preshutter reset row address
P1:0xD9	VSYNC_RISE_RCNT	0x00	RW	Bit[7:0]: vsync_rise_rcnt[7:0] Master or slave FSIN sync row address
P1:0xDA	VSYNC_RISE_RCNT	0x64	RW	Bit[3:0]: vsync_rise_rcnt[11:8] Master or slave FSIN sync row address
P1:0xDB	VSYNC_RISE_CCNT	0x00	RW	Bit[7:0]: exter_sync_dly_num[7:0] Master or slave FSIN sync column address
P1:0xDC	VSYNC_RISE_CCNT	0x8D	RW	Bit[2:0]: exter_sync_dly_num[10:8] Master or slave FSIN sync column address
P1:0xDE	EXTER_SYNC_OUT_WIDTH	0x01	RW	Bit[7:0]: exter_sync_out_width Master FSIN width

table 4-4 FSIN control registers (sheet 2 of 2)

address	register name	default value	R/W	description
P1:0xDF	EVSYNC_CTRL1	0x00	RW	Bit[7]: fsin_start_ctrl Bit[6]: fsin_pre_shutter_en Bit[5]: rstmode_nodelay Bit[4]: blank_mode_en Bit[3]: frame_start_mode Bit[2]: mipi_en_sel Bit[1]: fsin_inv Bit[0]: vsync_out_sel
P1:0xE2	EVSYNC_CTRL2	0x00	RW	Bit[0]: evsync_ctrl2

4.6 black level calibration (BLC)

The OS04J10 pixel array contains several optically black lines, which can be seen at the pixel array structure section. These lines are used to provide the data for black level calibration and further correction.

Ideally, the image sensor output without incident light illumination is zero, but even without considering the influence of the dark current, due to the presence of the offset of the readout circuit and the ADC itself, the final output is non-zero in the absence of illumination.

Black level calibration (BLC), is dark level calibration designed to eliminate the influence of the above offset, making the sensor imaging more in line with human perception. The usual practice for dark level calibration is to use an optical black pixel to count the offset under zero illumination and then subtract this offset from the value read by the active pixel.

4.7 one-time programmable (OTP) memory

The OS04J10 supports a maximum of 256 bytes of one-time programmable (OTP) memory to store chip identification and manufacturing information, and can be controlled through the SCCB

table 4-5 OTP memory usage

OTP address	default value	content
0x00	0x80	KSCG
0x01	0x04	tpm_otp_reg1
0x02	0xDD	tpm_otp_reg2
0x03	0xB9	tpm_otp_reg3
0x04	0x73	tpm_otp_reg4
0x05		for CP
0x06		for CP
...	...	...
0x16		for CP
0x17		for CP
0x18		OTP DPC info
0x19		OTP DPC info
...	...	...
0xFF		OTP DPC info

4.8 data compression algorithm

4.8.1 16-bit to 12-bit

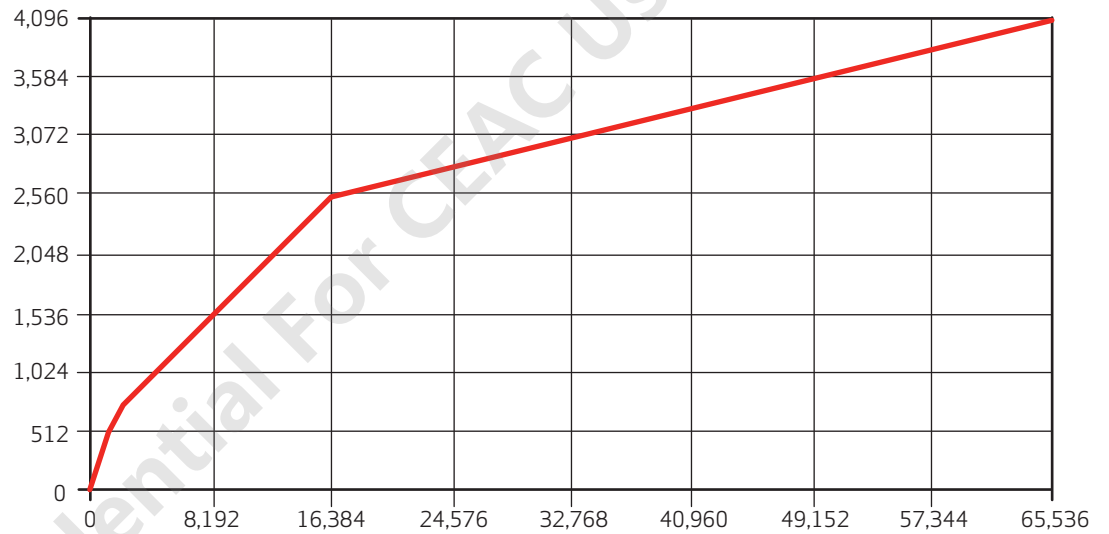
The OS04J10 offers data compression from 16-bit to 12-bit by providing a single piece-wise linear (PWL) method, with three options, utilizing different numbers of different fixed value knee points. The image data compression is enabled by setting register bit P7:0x35[4] = 1. The three different options can be selected via register bits P7:0x35[6:5] and are defined by the following formulas and curves shown in **figure 4-6** to **figure 4-8**.

4.8.1.1 option A compression – register bits P7:0x35[6:5] = 0

There are three fixed knee points, segmenting the value curve into four sections. The compression formula from 16-bit to 12-bit is given below for this particular case:

$$y_{out_12b} = \begin{cases} \frac{y_{in_16b}}{2}, & y_{in_16b} < 1024 \\ \frac{y_{in_16b}}{4} + 256, & 1024 \leq y_{in_16b} < 2048 \\ \frac{y_{in_16b}}{8} + 512, & 2048 \leq y_{in_16b} < 16384 \\ \frac{y_{in_16b}}{32} + 2048, & y_{in_16b} \geq 16384 \end{cases}$$

figure 4-6 16-bit to 12-bit PWL compression (option A)



The backend processor can decompress (option A) 12-bit data to 16-bit data using the following formula:

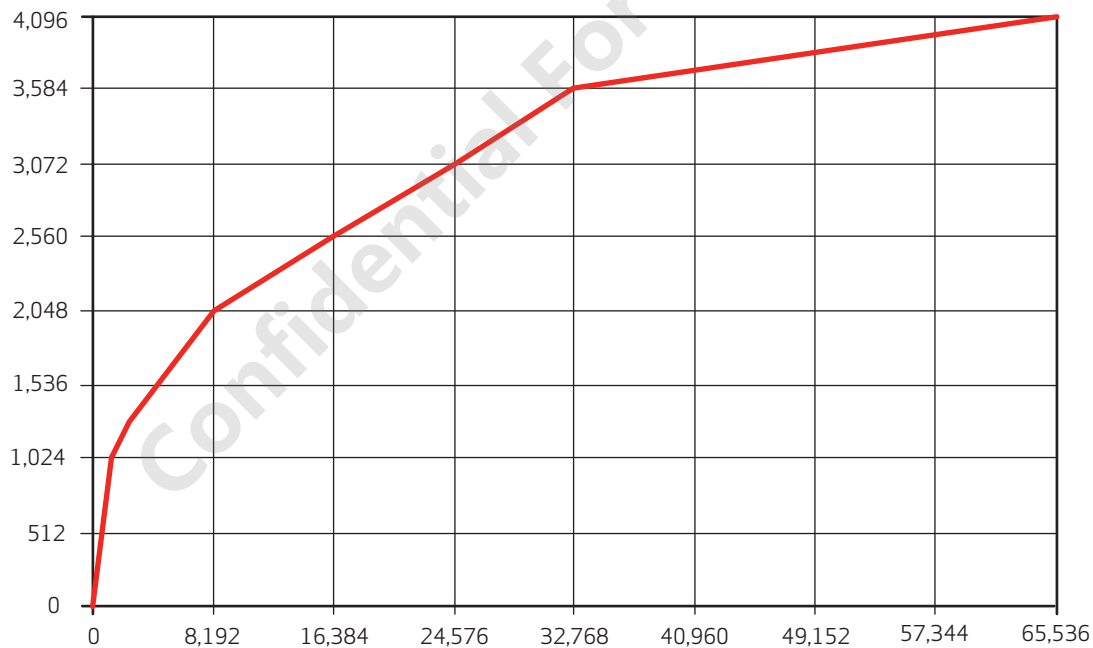
$$y_{out_16b} = \begin{cases} 2 \times y_{in_12b}, & y_{in_12b} < 512 \\ 4 \times (y_{in_12b} - 256), & 512 \leq y_{in_12b} < 768 \\ 8 \times (y_{in_12b} - 512), & 768 \leq y_{in_12b} < 2560 \\ 32 \times (y_{in_12b} - 2048), & y_{in_12b} \geq 2560 \end{cases}$$

4.8.1.2 option B compression – register bits P7:0x35[6:5] = 1

There are four fixed knee points, segmenting the value curve into five sections. The compression formula from 16-bit to 12-bit is given below for this particular case:

$$y_{out_{12b}} = \begin{cases} y_{in16b}, & y_{in16b} < 1024 \\ \frac{y_{in16b}}{4} + 768, & 1024 \leq y_{in16b} < 2048 \\ \frac{y_{in16b}}{8} + 1024, & 2048 \leq y_{in16b} < 8192 \\ \frac{y_{in16b}}{16} + 1536, & 8192 \leq y_{in16b} < 32768 \\ \frac{y_{in16b}}{64} + 3072, & y_{in16b} \geq 32768 \end{cases}$$

figure 4-7 16-bit to 12-bit PWL compression (option B)



The backend processor can decompress (option B) 12-bit data to 16-bit data using the following formula:

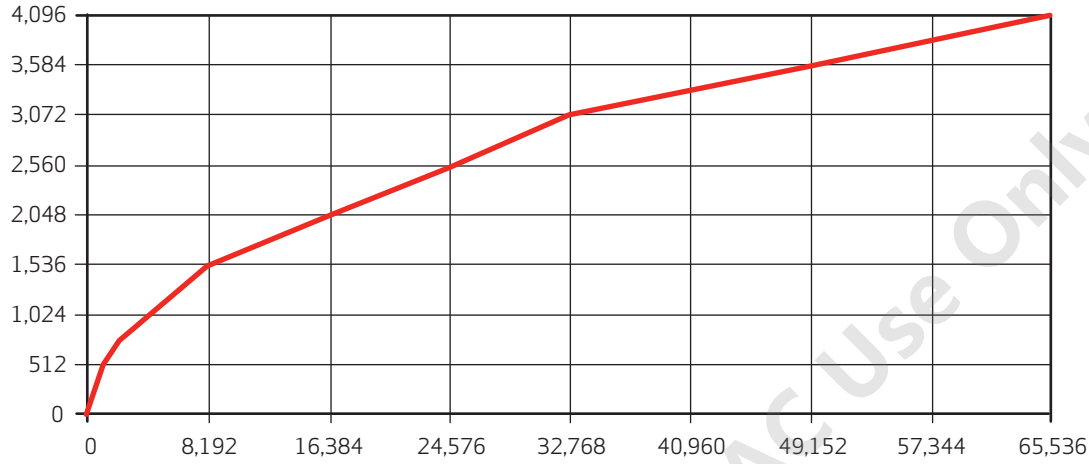
$$y_{out16b} = \begin{cases} y_{in12b}, & y_{in12b} < 1024 \\ 4 \times (y_{in12b} - 768), & 1024 \leq y_{in12b} < 1280 \\ 8 \times (y_{in12b} - 1024), & 1280 \leq y_{in12b} < 2048 \\ 16 \times (y_{in12b} - 1536), & 2048 \leq y_{in12b} < 3584 \\ 64 \times (y_{in12b} - 3072), & y_{in12b} \geq 3584 \end{cases}$$

4.8.1.3 option C compression – register bits P7:0x35[6:5] = 2

There are five fixed knee points, segmenting the value curve into six sections. The compression formula from 16-bit to 12-bit is given below for this particular case:

$$y_{out12b} = \begin{cases} y_{in16b}, & y_{in16b} < 256 \\ \frac{y_{in16b}}{2} + 128, & 256 \leq y_{in16b} < 512 \\ \frac{y_{in16b}}{4} + 256, & 512 \leq y_{in16b} < 2048 \\ \frac{y_{in16b}}{8} + 512, & 2048 \leq y_{in16b} < 8192 \\ \frac{y_{in16b}}{16} + 1024, & 8192 \leq y_{in16b} < 32768 \\ \frac{y_{in16b}}{32} + 2048, & y_{in16b} \geq 32768 \end{cases}$$

figure 4-8 16-bit to 12-bit PWL compression (option C)



The backend processor can decompress (option C) 12-bit data to 16-bit data using the following formula:

$$y_{out16b} = \begin{cases} y_{in12b}, & y_{in12b} < 256 \\ 2 \times (y_{in12b} - 128), & 256 \leq y_{in12b} < 384 \\ 4 \times (y_{in12b} - 256), & 384 \leq y_{in12b} < 768 \\ 8 \times (y_{in12b} - 512), & 768 \leq y_{in12b} < 1536 \\ 16 \times (y_{in12b} - 1024), & 1536 \leq y_{in12b} < 3072 \\ 32 \times (y_{in12b} - 2048), & y_{in12b} \geq 3072 \end{cases}$$

Option A is suitable in high light conditions, when losing one bit in the lower range is acceptable and the user wants to keep the maximum bits in the higher ranges. For low light conditions, option B is preferable. All the bits in the lower range are maintained, while only losing more bits in the higher range where it is more acceptable. Option C is a hybrid of A and B. Option C retains all the bits in the very low range and maintains the maximum bits in the higher ranges.

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5 register tables

5.1 module name and address range

table 5-1 module name and address range

module name	address range
system control	P0:0x02~P0:0xEA
sensor control	P1:0x04~P1:0xF0, P2:0x36~P2:0x9C
ISP	P3:0x00~P3:0xEA
dynamic DPC long frame	P4:0x00~P4:0x22
dynamic DPC short frame	P4:0x80~P4:0x9B
BLC	P5:0x00~P5:0xDB
OTP DPC	P6:0x04~P6:0x2B
DCG combine	P7:0x00~P7:0x38
OTP	P9:0x80~P9:0xC3
page selection	0xFD

5.2 system control [P0:0x02 ~ P0:0xEA]

table 5-2 system registers (sheet 1 of 7)

address	register name	default value	R/W	description
P0:0x02	CHIP_ID	0x53	R	Bit[7:0]: chip_id[31:24]
P0:0x03	CHIP_ID	0x04	R	Bit[7:0]: chip_id[23:16]
P0:0x04	CHIP_ID	0x4A	R	Bit[7:0]: chip_id[15:8]
P0:0x05	CHIP_ID	0x10	R	Bit[7:0]: chip_id[7:0]
P0:0x06	RSVD	—	—	Reserved

table 5-2 system registers (sheet 2 of 7)

address	register name	default value	R/W	description
P0:0x07	IO_SET01_ADDR	0x00	RW	Bit[7]: vsync_sel 0: vsyncL_isp 1: vsyncS_isp Bit[6]: vsync_inv_en Bit[5:4]: strobe_ds Bit[3:2]: fsin_ds Bit[1:0]: vsync_ds
P0:0x08	IO_SET02_ADDR	0x2A	RW	Bit[7]: strobe_oen_direct Bit[6]: vsync_oen_direct Bit[5]: strobe_oen Bit[4]: strobe_ie Bit[3]: fsin_oen Bit[2]: fsin_ie Bit[1]: vsync_oen Bit[0]: vsync_ie
P0:0x09	P0_COMC00	0x00	RW	Bit[7:4]: cnt_psv_sel Bit[3]: Reserved Bit[2]: pwd_pll2 Bit[1]: pwd_pll1 Bit[0]: pwd_asp
P0:0x0A	P0_COMC01	0x40	RW	Bit[7]: Not used Bit[6]: cnt_test_en Bit[5]: cnt_psv_en Bit[4]: dac_clk_div_sel Bit[3]: bank_close_psv_en Bit[2]: cnt_clk_div_sel Bit[1]: col_sf_en Bit[0]: pix_nb_cas_en
P0:0x0B	P0_COMC02	0xD4	RW	Bit[7:4]: col_comp1_bias Bit[3:2]: col_sf_bias Bit[1]: pix_idle_en Bit[0]: col_en_samp_clk_phase_sel
P0:0x0C	P0_COMC03	0x00	RW	Bit[7]: ramp_idle_en Bit[6:4]: readout_dly_sel[2:0] Bit[3]: pwd_bsun_sig_lcg Bit[2]: pwd_bsun_sig_hcg Bit[1]: pwd_bsun_rst_lcg Bit[0]: pwd_bsun_rst_hcg
P0:0x0D	P0_COMC04	0x40	RW	Bit[7:4]: Not used Bit[3:2]: col_en_samp_clk_div_sel Bit[1]: cp_clk_sel Bit[0]: cnt_clk_test_en
P0:0x0E~ P0:0x0F	RSVD	–	–	Reserved
P0:0x10	P0_COMC05	0xCC	RW	Bit[7:0]: mpll_dsm[15:8]

table 5-2 system registers (sheet 3 of 7)

address	register name	default value	R/W	description
P0:0x11	P0_COMC06	0x00	RW	Bit[7:0]: mpll_dsm[7:0]
P0:0x12	P0_COMC07	0x42	RW	Bit[7:4]: mpll_dsm[19:16] Bit[3]: dppll_cp_clk_disab_sel Bit[2]: mpll_rst_sync_en Bit[1]: mpll_sscg_cntstep_x3 Bit[0]: mpll_sscg_en
P0:0x13	P0_COMC08	0x06	RW	Bit[7]: mpll_bp_vco_sync Bit[6]: mpll_dither_en Bit[5]: mpll_div_pwd Bit[4]: mpll_frac_en Bit[3]: mpll_loop_pwd Bit[2:0]: mpll_sscg_cntck
P0:0x14	P0_COMC09	0x1E	RW	Bit[7]: vdd_clamp_dn_en Bit[6]: vdd_clamp_up_en Bit[5:4]: mpll_cp_clk_post_sel Bit[3:2]: mpll_cp_clk_pre_sel Bit[1]: mpll_cp_clk_disab_sel Bit[0]: mpll_pclk_disab_sel
P0:0x15	P0_COMC0A	0x01	RW	Bit[7:6]: dppll_cp_sel Bit[5]: dppll_div_pwd Bit[4]: dppll_band_sel Bit[3]: dppll_en_exti_sel Bit[2]: dppll_predivp_sel Bit[1]: dppll_rst_sync_sel Bit[0]: dppll_vcoby2_sel
P0:0x16	DPLL_NC_SEL	0x00	RW	Bit[7:2]: Not used Bit[1:0]: dppll_nc_sel[9:8]
P0:0x17	DPLL_NC_SEL	0x7E	RW	Bit[7:0]: dppll_nc_sel[7:0]
P0:0x18	DPLL_MC_SEL	0x05	RW	Bit[7:3]: Not used Bit[2:0]: dppll_mc_sel
P0:0x19	P0_COMC0B	0x60	RW	Bit[7]: dppll_row_clk_disab_sel Bit[6:4]: dppll_row_clk_post_sel Bit[3:2]: dppll_cnt_clk_sel Bit[1]: Reserved Bit[0]: dppll_byp_sel
P0:0x1A	P0_COMC0C	0x65	RW	Bit[7]: dppll_cp_clk_pre_sel Bit[6]: dppll_pclk_disab_sel Bit[5:4]: dppll_pclk_post_sel Bit[3:2]: dppll_pclk_pre_sel Bit[1:0]: dppll_row_clk_pre_sel
P0:0x1B	P0_COMC12	0x21	RW	Bit[7:4]: mpll_cp_sel Bit[3:0]: mpll_sscg_cntstep

table 5-2 system registers (sheet 4 of 7)

address	register name	default value	R/W	description
P0:0x1C	MPLL_NC_SEL	0x00	RW	Bit[7:2]: Not used Bit[1:0]: mpll_nc_sel[9:8]
P0:0x1D	MPLL_NC_SEL	0xF0	RW	Bit[7:0]: mpll_nc_sel[7:0]
P0:0x1E	MPLL_MC_SEL	0x05	RW	Bit[7:3]: Not used Bit[2:0]: mpll_mc_sel
P0:0x1F	MPLL_COMC02	0x21	RW	Bit[7:6]: Not used Bit[5:4]: mpll_pclk_post_sel Bit[3]: mpll_bclk_sel Bit[2:0]: mpll_pclk_pre_sel
P0:0x20	MPLL_COMC03	0x01	RW	Bit[7:6]: Not used Bit[5]: mpll_byp_sel Bit[4]: mpll_predivp_sel Bit[3]: Reserved Bit[2:0]: mpll_phy_clk_sel
P0:0x21	P0_COMC15	0x27	RW	Bit[7]: Not used Bit[6:4]: mpll_opa_sel Bit[3]: Reserved Bit[2:0]: dpll_cp_clk_post_sel
P0:0x22~ P0:0x23	RSVD	—	—	Reserved
P0:0x24	P0_COMC0D	0x40	RW	Bit[7]: mipi_valid_sel Bit[6]: mipi_vb_psv_byp Bit[5:4]: mipi_vof_sel[1:0] Bit[3]: mipi_oterm_en Bit[2:0]: mipi_vcm_sel
P0:0x25	P0_COMC0E	0x01	RW	Bit[7:6]: mipi_ck_skew_sel Bit[5]: mipi_eq_en Bit[4]: mipi_hsen_skew_sel Bit[3]: mipi_lp_rtime_dis Bit[2]: mipi_lp_sr_sel Bit[1:0]: mipi_lptx_sel
P0:0x26	P0_COMC0F	0x00	RW	Bit[7:6]: mipi_d0_skew_sel Bit[5:4]: mipi_d1_skew_sel Bit[3:2]: mipi_d2_skew_sel Bit[1:0]: mipi_d3_skew_sel
P0:0x27	CLK_MODE	0x07	RW	Bit[7:3]: Not used Bit[2]: dclk_otp_dpc_gating_en Bit[1]: dclk_chn_gating_en Bit[0]: dclk2x_gating_en

table 5-2 system registers (sheet 5 of 7)

address	register name	default value	R/W	description
P0:0x28	CLK_MODE	0xFE	RW	Bit[7]: Not used Bit[6]: bclk2x_gating_en Bit[5]: dclkin_gating_en Bit[4]: dclk_dpc_gating_en Bit[3]: dclk_blc_gating_en Bit[2]: dclk_mipi_gating_en Bit[1]: bclk_gating_en Bit[0]: pclk_in_sel 0: pclk_in 1: EXTCLK
P0:0x29	CLK_MODE	0x20		Bit[7:6]: Not used Bit[5:4]: row_clk_sel Bit[3]: Reserved Bit[2]: dclk_ctrl Bit[1:0]: timer_clk_ctrl
P0:0x2A~ P0:0x2B	RSVD	–	–	Reserved
P0:0x2C	MIPI_EN	0x00	RW	Bit[7:1]: Not used Bit[0]: mipi_en
P0:0x2D	R_CLK_TRAIL	0x07	RW	Bit[7:0]: r_clk_trail
P0:0x2E	R_EXIT	0x04	RW	Bit[7:0]: r_exit
P0:0x2F	R_CLK_PREPARE	0x04	RW	Bit[7:0]: r_clk_prepare
P0:0x30	H_SIZE_MIPI	0x0A	RW	Bit[7:2]: Not used Bit[3:0]: h_size_mipi[11:8]
P0:0x31	H_SIZE_MIPI	0x08	RW	Bit[7:0]: h_size_mipi[7:0]
P0:0x32	V_SIZE_MIPI	0x05	RW	Bit[7:3]: Not used Bit[2:0]: v_size_mipi[10:8]
P0:0x33	V_SIZE_MIPI	0xA8	RW	Bit[7:0]: v_size_mipi[7:0]
P0:0x34	HS_MODE	0x00	RW	Bit[7:2]: Not used Bit[1]: hs_mode_vf Bit[0]: hs_mode
P0:0x35	R_CLK_POST	0x0E	RW	Bit[7:0]: r_clk_post
P0:0x36	R_LPX_CK	0x07	RW	Bit[7:0]: r_lpx_ck
P0:0x37	R_HS_PREPARE	0x06	RW	Bit[7:0]: r_hs_prepare
P0:0x38	R_HS_ZERO	0x1A	RW	Bit[7:0]: r_hs_zero
P0:0x39	DATA_ID_REG	0x2B	RW	Bit[7:0]: data_id_reg
P0:0x3A	R_HS_TRAIL	0x08	RW	Bit[7:0]: r_hs_trail
P0:0x3B	R_CLK_ZERO	0x22	RW	Bit[7:0]: r_clk_zero

table 5-2 system registers (sheet 6 of 7)

address	register name	default value	R/W	description
P0:0x3C	R_LPX_DAT	0x07	RW	Bit[7:0]: r_lpx_dat
P0:0x3D	TX_SPEED_AREA_SEL	0x03	RW	Bit[7:3]: Not used Bit[2:0]: tx_speed_area_sel
P0:0x3E	R_INIT	0x0B	RW	Bit[7:0]: r_init[15:8]
P0:0x3F	R_INIT	0x40	RW	Bit[7:0]: r_init[7:0]
P0:0x40	R_WAKEUP	0x00	RW	Bit[7:2]: Not used Bit[1:0]: r_wakeup[17:16]
P0:0x41	R_WAKEUP	0x86	RW	Bit[7:0]: r_wakeup[15:8]
P0:0x42	R_WAKEUP	0x88	RW	Bit[7:0]: r_wakeup[7:0]
P0:0x43	DC_TEST_LP_LK	0x3F	RW	Bit[7]: dc_test_hsd_c_en Bit[6]: dc_test_hsd_d_en Bit[5:4]: dc_test_lp_ck_en Bit[3:2]: dc_test_lp_d1_en Bit[1:0]: Reserved
P0:0x44	DC_TEST_DATA_HS	0xFF	RW	Bit[7:0]: dc_test_data_hs
P0:0x45	FRAME_END_DLY	0x65	RW	Bit[7:0]: frame_end_dly
P0:0x46	MIPI_HSYNC_NUM	0x05	RW	Bit[7:0]: mipi_hsync_num
P0:0x47	MIPI_CTRL	0x20	RW	Bit[7:6]: Not used Bit[5]: mp_double Bit[4]: pause_ck Bit[3]: ls_mode Bit[2]: Init Bit[1]: ulp_mode Bit[0]: Testmode
P0:0x48~ P0:0x49	RSVD	–	–	Reserved
P0:0x4A	HBLANK_READ	–	R	Bit[7:6]: Not used Bit[5:0]: hblank_read[13:8]
P0:0x4B	HBLANK_READ	–	R	Bit[7:0]: hblank_read[7:0]
P0:0x4C	MIPI_LS_CK	0x01	RW	Bit[7:4]: Not used Bit[3:2]: mipi_ls_start_num Bit[1:0]: mipi_ck_data
P0:0x4D	MIPI_CLK_INV	0x01	RW	Bit[7:1]: Not used Bit[0]: mipi_clk_inv
P0:0x4E~ P0:0x51	RSVD	–	–	Reserved
P0:0x52	CLK_TRAIL_MODE	0x01	RW	Bit[7:1]: Not used Bit[0]: clk_trail_mode

table 5-2 system registers (sheet 7 of 7)

address	register name	default value	R/W	description
P0:0x53	MODE_10BIT	0x01	RW	Bit[7:1]: Not used Bit[0]: mode_10bit 0: 12-bit 1: 10-bit
P0:0x54	RSVD	–	–	Reserved
P0:0x55	R_CLK_TIME	0x00	RW	Bit[7]: Not used Bit[6:0]: clk_time
P0:0x56	R_DATA_TIME	0x00	RW	Bit[7:6]: Not used Bit[5:0]: data_time
P0:0x57	P0_COMC16	0x00	RW	Bit[7:5]: Not used Bit[4]: soft_otp_rstb_buf Bit[3]: Reserved Bit[2:0]: comp_hb_bank_sel
P0:0x58	P0_COMC10	0xB0	RW	Bit[7:4]: mipi_ck_drv_sel Bit[3]: mipi_ck_test_en Bit[2]: mipi_d1_lane_dis Bit[1]: mipi_d2_lane_dis Bit[0]: mipi_d3_lane_dis
P0:0x59	P0_COMC11	0x03	RW	Bit[7:2]: Not used Bit[1]: mipi_pwd_sel Bit[0]: mipi_p2s_pwd
P0:0x5A	P0_COMC13	0x0B	RW	Bit[7]: Not used Bit[6:4]: col_clk_dly_sel Bit[3:0]: mipi_data_drv_sel
P0:0x5B~ P0:0xE9	RSVD	–	–	Reserved
P0:0xEA	MIPE_REG_UPDATE	0x00	RW	Bit[7:2]: Not used Bit[1:0]: mipi_reg_update

5.3 sensor control [P1:0x04 - P1:0xF0, P2:0x36 - P2:0x9C]

table 5-3 sensor control registers (sheet 1 of 8)

address	register name	default value	R/W	description
P1:0x04	VSTART	0x00	RW	Bit[7:3]: Not used Bit[2:0]: Analog window vstart[10:8]
P1:0x05	VSTART	0x20	RW	Bit[7:0]: Analog window vstart[7:0]

table 5-3 sensor control registers (sheet 2 of 8)

address	register name	default value	R/W	description
P1:0x06~ P1:0x07	RSVD	—	—	Reserved
P1:0x08	VSIZE	0x05	RW	Bit[7:3]: Not used Bit[2:0]: Analog window vsize[10:8]
P1:0x09	VSIZE	0xA8	RW	Bit[7:0]: Analog window vsize[7:0]
P1:0x0A~ P1:0x0D	RSVD	—	—	Reserved
P1:0x0E	TEXP	0x00	RW	Bit[7:0]: Texp[15:8]
P1:0x0F	TEXP	0x01	RW	Bit[7:0]: Texp[7:0]
P1:0x10~ P1:0x11	RSVD	—	—	Reserved
P1:0x12	FLIP/MIRROR	0x00	RW	Bit[7:4]: Not used Bit[3]: Preserve Bayer pattern on mirror Bit[2]: Preserve Bayer pattern on flip Bit[1:0]: Image mirror/flip 00: Normal 01: Vertical flip 10: Horizontal mirror 11: Mirror and flip
P1:0x13	RSVD	—	—	Reserved
P1:0x14	VBLANK	0x00	RW	Bit[7:0]: Vblank[15:8] Mix value is 4
P1:0x15	VBLANK	0x04	RW	Bit[7:0]: Vblank[7:0] Mix value is 4
P1:0x16	TEXP_LSB0	0x00	RW	Bit[7:1]: Not used Bit[0]: texp_lsb0
P1:0x17	VBLANK_LSB0	0x00	RW	Bit[7:1]: Not used Bit[0]: vblank_lsb0
P1:0x18	FPSHIP	0x00	RW	Bit[7:1]: Not used Bit[0]: Fpship
P1:0x19	EXP_MODE	0x00	RW	Bit[7:3]: Not used Bit[2:0]: exp_mode 000: linear 001: DCG 010: HDR 2 stagger 110: HDR 2 stagger1 Others: Not used
P1:0x1A	TEXP_S	0x00	RW	Bit[7:0]: texp_s[15:8]
P1:0x1B	TEXP_S	0x01	RW	Bit[7:0]: texp_s[7:0]

table 5-3 sensor control registers (sheet 3 of 8)

address	register name	default value	R/W	description
P1:0x1C~ P1:0x1E	RSVD	–	–	Reserved
P1:0x1F	DIG_GAIN	0x00	RW	Bit[7:3]: Not used Bit[2:0]: dig_gain[9:7]
P1:0x20	DIG_GAIN	0x40	RW	Bit[7:1]: dig_gain[6:0] Bit[0]: Not used
P1:0x21	RAMP_PGA_GAIN	0x00	RW	Bit[7:0]: ramp_pga_gain
P1:0x22	RPC_EN	0x01	RW	Bit[7:2]: Not used Bit[1]: rpc_map_sel Bit[0]: rpc_en
P1:0x23	NOT USED	–	–	Not used
P1:0x24	RPC	0x10	RW	Bit[7:0]: RPC[7:0]
P1:0x25	HDRGAP0	0x01	RW	Bit[7:6]: Not used Bit[5:0]: Hdrgap0[13:8]
P1:0x26	HDRGAP0	0x10	RW	Bit[7:0]: Hdrgap0[7:0]
P1:0x27	HDRGAP1	0x01	RW	Bit[7:6]: Not used Bit[5:0]: Hdrgap1[13:8]
P1:0x28	HDRGAP1	0x10	RW	Bit[7:0]: Hdrgap1[7:0]
P1:0x29	RSVD	–	–	Reserved
P1:0x2A	ROWTIME	0x01	R	Bit[7:6]: Not used Bit[5:0]: Rowtime[13:8]
P1:0x2B	ROWTIME	0x10	R	Bit[7:0]: Rowtime[7:0]
P1:0x2C	ROWTIME_S	0x01	R	Bit[7:6]: Not used Bit[5:0]: rowtime_s[13:8]
P1:0x2D	ROWTIME_S	0x10	R	Bit[7:0]: rowtime_s[7:0]
P1:0x2E	RSVD	–	–	Reserved
P1:0x2F	FRMTIME	0x04	R	Bit[7:0]: Frame time[15:8]
P1:0x30	FRMTIME	0x55	R	Bit[7:0]: Frame time[7:0]
P1:0x31	FRMTIME_DELTA	0x10	RW	Bit[7:5]: Not used Bit[4:0]: Frame time delta[4:0]
P1:0x32	VBINNING_EN	0x00	RW	Bit[7:2]: Not used Bit[1]: Enable mono binning Bit[0]: Enable color binning Must be configured at same time as register P3:0x0E for color binning

table 5-3 sensor control registers (sheet 4 of 8)

address	register name	default value	R/W	description
P1:0x33	COMMC	0x38	RW	Bit[7]: gain_test Bit[6]: Not used Bit[5]: old_idx2_en Bit[4]: offset_row_adr_replace Bit[3]: darkrow_output_en (only for FSYNC) Bit[2]: double_shutter_off Bit[1]: dcg_rpc_sel 0: 1/2/4/8/16x 1: RPC Bit[0]: Reserved
P1:0x34	ANA_V_DUMMY	0x20	RW	Bit[7:0]: ana_v_dummy
P1:0x35~ P1:0x3A	RSVD	—	—	Reserved
P1:0x3B	DUMMY_ROW1	0x0A	RW	Bit[7:0]: dummy_row1
P1:0x3C	DUMMY_ROW2	0x0A	RW	Bit[7:0]: dummy_row2
P1:0x3D	VB_LP_MODE	0x06	RW	Bit[7:3]: Reserved Bit[2]: vb_psv_pl_en Bit[1]: vb_psv_exp_en Bit[0]: vb_lp_mode
P1:0x3E~ P1:0x3F	RSVD	—	—	Reserved
P1:0x40	ULP_EXP_LIMIT	0x05	RW	Bit[7:0]: ulp_exp_limit[15:8]
P1:0x41	ULP_EXP_LIMIT	0xC8	RW	Bit[7:0]: ulp_exp_limit[7:0]
P1:0x42	ULP_FL_LIMIT	0x05	RW	Bit[7:0]: ulp_fl_limit[15:8]
P1:0x43	ULP_FL_LIMIT	0xF0	RW	Bit[7:0]: ulp_fl_limit[7:0]
P1:0x44	VSNC_SC_DLY	0x07	RW	Bit[7:0]: vsync_sc_dly[7:0]
P1:0x45~ P1:0x4D	RSVD	—	—	Reserved
P1:0x4E	DEL_FRM_MODE	0x01	RW	Bit[7:1]: Reserved Bit[0]: del_mirror_change
P1:0x4F	DEL_FRM_MODE	0x41	RW	Bit[7]: del_fsin_frm Bit[6]: del_updown_change Bit[5]: del_exp_rpc_change Bit[4]: del_startexp Bit[3:0]: del_frm_num
P1:0x50~ P1:0xCF	RSVD	—	—	Reserved

table 5-3 sensor control registers (sheet 5 of 8)

address	register name	default value	R/W	description
P1:0xD0	BLC_OB1	0xD2	RW	Bit[7:4]: Ob1 end address Bit[3:0]: Ob1 start address
P1:0xD1	BLC_OB2	0x00	RW	Bit[7:4]: Ob2 end address[3:0] Bit[3:0]: Ob2 start address
P1:0xD2	BLC_OB2	0xB4	RW	Bit[7:1]: Not used Bit[0]: Ob2 end address[4]
P1:0xD3	EVSYNC_CTRL	0x00	RW	Bit[7]: exter_sync_inv Bit[6]: Not used Bit[5]: rst_colrow_en Bit[4]: slave_preshutter_en Bit[3]: Not used Bit[2]: sync_no_wait_en Bit[1]: exter_slave_en Bit[0]: exter_master_en
P1:0xD4	RSVD	–	–	Reserved
P1:0xD5	FSIN_OUT_FORMAT	0x00	RW	Bit[7:1]: Not used Bit[0]: fsin_out_format 0: fsin_out 1: vsync_isp
P1:0xD6	EXTER_SYNC_FRAME_NUM	0x01	RW	Bit[7:0]: exter_sync_frame_num
P1:0xD7	SLAVE_SHUTTER_PERIOD	0x00	RW	Bit[7:0]: Slave shutter period[7:0] Slave preshutter reset row address
P1:0xD8	SLAVE_SHUTTER_PERIOD	0x01	RW	Bit[7:4]: Not used Bit[3:0]: Slave shutter period[11:8] Slave preshutter reset row address
P1:0xD9	VSYNC_RISE_RCNT	0x00	RW	Bit[7:0]: vsync_rise_rcnt[7:0] Master or slave FSIN sync row address
P1:0xDA	VSYNC_RISE_RCNT	0x64	RW	Bit[7:4]: Not used Bit[3:0]: vsync_rise_rcnt[11:8] Master or slave FSIN sync row address
P1:0xDB	VSYNC_RISE_CCNT	0x00	RW	Bit[7:0]: exter_sync_dly_num[7:0] Master or slave FSIN sync column address
P1:0xDC	VSYNC_RISE_CCNT	0x8D	RW	Bit[7:3]: Not used Bit[2:0]: exter_sync_dly_num[10:8] Master or slave FSIN sync column address

table 5-3 sensor control registers (sheet 6 of 8)

address	register name	default value	R/W	description
P1:0xDD	RSVD	–	–	Reserved
P1:0xDE	EXTER_SYNC_OUT_WIDTH	0x01	RW	Bit[7:0]: exter_sync_out_width Master FSIN width
P1:0xDF	EVSYNC_CTRL1	0x00	RW	Bit[7]: fsin_start_ctrl Bit[6]: fsin_pre_shutter_en Bit[5]: rstmode_nodelay Bit[4]: blank_mode_en Bit[3]: frame_start_mode Bit[2]: mipi_en_sel Bit[1]: fsin_inv Bit[0]: vsync_out_sel
P1:0xE0~ P1:0xE1	RSVD	–	–	Reserved
P1:0xE2	EVSYNC_CTRL2	0x00	RW	Bit[7:1]: Not used Bit[0]: evysnc_ctrl2
P1:0xE3	RSVD	–	–	Reserved
P1:0xE4	DIG_GAIN_S	0x00	RW	Bit[7:3]: Not used Bit[2:0]: dig_gain_s[9:7]
P1:0xE5	DIG_GAIN	0x40	RW	Bit[7:1]: dig_gain_s[6:0] Bit[0]: Not used
P1:0xE6	NOT USED	–	–	Not used
P1:0xE7	RPC_S	0x10	RW	Bit[7:0]: rpc_s[7:0]
P1:0xE8	KSCG	0x40	RW	Bit[7:0]: KSCG
P1:0xE9	KSCG_S	0x40	RW	Bit[7:0]: kscg_s
P1:0xEA	EXP_OFFSET_REG_EN	0x00	RW	Bit[7:0]: exp_offset_reg_en
P1:0xEB	EXP_OFFSET_REG	0x5D	RW	Bit[7:0]: exp_offset_reg
P1:0xEC	EXP_OFFSET	–	R	Bit[7:0]: exp_offset
P1:0xED	SCG_EN	0x01	RW	Bit[7:1]: Not used Bit[0]: scg_en
P1:0xEE	SCG_S_EN	0x01	RW	Bit[7:1]: Not used Bit[0]: scg_s_en
P1:0xEF	HDR_DUMMY	0x00	RW	Bit[7:1]: Not used Bit[0]: hdr_dummy
P1:0xF0	HDR_LS_SEL	0x03	RW	Bit[7:2]: Not used Bit[1]: HDR mode long frame output enable Bit[0]: HDR mode short frame output enable

table 5-3 sensor control registers (sheet 7 of 8)

address	register name	default value	R/W	description
P2:0x36	R_STG_VC	0x21	RW	Bit[7:0]: r_stg_vc
P2:0x37~ P2:0x7F	RSVD	–	–	Reserved
P2:0x80	R0	0x00	RW	Bit[7:0]: r_start_size[15:8]
P2:0x81	R1	0x30	RW	Bit[7:0]: r_start_size[7:0]
P2:0x82	R2	0x02	RW	Bit[7:4]: r_rm Bit[3]: r_test1 Bit[2]: r_bist_sel Bit[1]: r_frame_rst_en Bit[0]: r_ram_byp
P2:0x83	R3	0x01	RW	Bit[7:4]: r_start_opt Bit[3]: r_fix_hsize Bit[2]: r_fo_rd_en_wr_cnd Bit[1]: r_rme Bit[0]: man_start_mode
P2:0x84	R4	0x00	RW	Bit[7:4]: Reserved Bit[3]: r_sram_hfq_mode Bit[2]: r_fo_mrr_4x_fix_dis Bit[1]: r_vfifo_mirror_dis Bit[0]: r_mirr_fi_swap_dis
P2:0x85	R5	0x03	RW	Bit[7]: Reserved Bit[6]: r_rd_sccb_bit_sel Bit[5]: r_vfifo_rd_sccb_en Bit[4]: r_sram_always_on Bit[3:0]: r_rdy_low
P2:0x86~ P2:0x88	RSVD	–	–	Reserved
P2:0x89	VFIFO_BIST_EN	0x00	RW	Bit[7:1]: Reserved Bit[0]: vfifo_bist_en
P2:0x8A	VFIFO_BIST_FLAG	–	R	Bit[7:2]: Reserved Bit[1]: error_flag_vfifo Bit[0]: test_done_vfifo
P2:0x8B~ P2:0x8F	RSVD	–	–	Reserved
P2:0x90	SHUTTER_MODE	0x00	RW	Bit[7:3]: Reserved Bit[2:0]: Select shutter mode 000: Rolling shutter Others: Global reset
P2:0x91	DUMMY_ROW3	0x02	RW	Bit[7:0]: dummy_row3
P2:0x92	DUMMY_ROW4	0x03	RW	Bit[7:0]: dummy_row4

table 5-3 sensor control registers (sheet 8 of 8)

address	register name	default value	R/W	description
P2:0x93	BULB_START_EXP	0x00	RW	Bit[7:1]: Reserved Bit[0]: bulb_start_exp
P2:0x94	GLOBAL_RESET_DELAY1	–	R	Bit[7:2]: Reserved Bit[1:0]: global_reset_delay1[9:8]
P2:0x95	GLOBAL_RESET_DELAY1	–	R	Bit[7:0]: global_reset_delay1[7:0]
P2:0x96	GLOBAL_RESET_EXP_PSV_EN	0x01	RW	Bit[7:1]: Reserved Bit[0]: global_reset_exp_psv_en
P2:0x97	STROBE_FILTER_MODE	0x00	RW	Bit[7:2]: Reserved Bit[1:0]: strobe_filter_mode
P2:0x98	STROBE_ADD_EXP	0x00	RW	Bit[7:0]: strobe_add_exp[15:8]
P2:0x99	STROBE_ADD_EXP	0x20	RW	Bit[7:0]: strobe_add_exp[7:0]
P2:0x9A	STROBE_CTRL	0x00	RW	Bit[7]: strobe_req Bit[6]: strobe_inv Bit[5:4]: xeon_width Bit[3]: strobe_level_en Bit[2]: led3_en Bit[1]: led12_en Bit[0]: xeon_en
P2:0x9B	STROBE_DEL_NUM	0x00	RW	Bit[7:0]: strobe_del_num
P2:0x9C	STROBE_LEVEL	0x00	RW	Bit[7:1]: Reserved Bit[0]: strobe_level

5.4 ISP [P3:0x00 ~ P3:0xEA]

table 5-4 ISP registers (sheet 1 of 4)

address	register name	default value	R/W	description
P3:0x00	HSTART_CIS	0x00	RW	Bit[7:4]: Reserved Bit[3:0]: hstart_cis[11:8]
P3:0x01	HSTART_CIS	0x00	RW	Bit[7:0]: hstart_cis[7:0]
P3:0x02	HSIZE_CIS	0x0A	RW	Bit[7:4]: Reserved Bit[3:0]: hsize_cis[11:8]
P3:0x03	HSIZE_CIS	0x08	RW	Bit[7:0]: hsize_cis[7:0]
P3:0x04	VSTART_CIS	0x00	RW	Bit[7:3]: Reserved Bit[2:0]: vstart_cis[10:8]

table 5-4 ISP registers (sheet 2 of 4)

address	register name	default value	R/W	description
P3:0x05	VSTART_CIS	0x20	RW	Bit[7:0]: vstart_cis[7:0]
P3:0x06	VSIZE_CIS	0x05	RW	Bit[7:3]: Reserved Bit[2:0]: vsize_cis[10:8]
P3:0x07	VSIZE_CIS	0xA8	RW	Bit[7:0]: vsize_cis[7:0]
P3:0x08~ P3:0x0B	RSVD	–	–	Reserved
P3:0x0C	VPRE_CIS	0x02	RW	Bit[7:0]: vpre_cis
P3:0x0D	CH_OFFSET	0x20	RW	Bit[7:0]: ch_offset
P3:0x0E	BINNING_EN	0x00	RW	Bit[7:2]: Reserved Bit[1]: Enable column binning average Bit[0]: Enable binning Must be configured at same time as register P1:0x32 for color binning
P3:0x0F	CH_BIST1	0x00	RW	Bit[7:3]: Reserved Bit[2]: Error flag (read only) Bit[1]: Test done (read only) Bit[0]: Enable test
P3:0x10	CH_BIST2	0x00	RW	Bit[7:3]: Reserved Bit[2]: Error flag (read only) Bit[1]: Test done (read only) Bit[0]: Enable test
P3:0x11	DSP_BYPASS	0x00	RW	Bit[7:1]: Reserved Bit[0]: DSP bypass
P3:0x12	BLK_LN_NUM	0x10	RW	Bit[7:5]: Reserved Bit[4:0]: BLK lane number
P3:0x13	ZERO_LN_NUM	0x10	RW	Bit[7:6]: Reserved Bit[5:0]: Zero lane number
P3:0x14	BL_POSITION_SET	0x02	RW	Bit[7:2]: Reserved Bit[1:0]: bl_position_set
P3:0x15	BL_POSITION_SET2	0x03	RW	Bit[7:2]: Reserved Bit[1:0]: bl_position_set2
P3:0x16	HSTART_WIN	0x00	RW	Bit[7:4]: Reserved Bit[3:0]: hstart_win[11:8]
P3:0x17	HSTART_WIN	0x00	RW	Bit[7:0]: hstart_win[7:0]
P3:0x18	HSIZE_WIN	0x0A	RW	Bit[7:4]: Reserved Bit[3:0]: hsize_win[11:8]
P3:0x19	HSIZE_WIN	0x08	RW	Bit[7:0]: hsize_win[7:0]

table 5-4 ISP registers (sheet 3 of 4)

address	register name	default value	R/W	description
P3:0x1A	VSTART_WIN	0x00	RW	Bit[7:3]: Reserved Bit[2:0]: vstart_win[10:8]
P3:0x1B	VSTART_WIN	0x00	RW	Bit[7:0]: vstart_win[7:0]
P3:0x1C	VSIZE_WIN	0x05	RW	Bit[7:3]: Reserved Bit[2:0]: vsize_win[10:8]
P3:0x1D	VSIZE_WIN	0xA8	RW	Bit[7:0]: vsize_win[7:0]
P3:0x1E~ P3:0x99	RSVD	—	—	Reserved
P3:0x9A	OTP_DPC	0x00	RW	Bit[7:2]: Reserved Bit[1]: otpdpc_en_s Bit[0]: otpdpc_en_l
P3:0x9B	OUTMODE1	0x00	RW	Bit[7:3]: Reserved Bit[2]: dis_ck_gt_otp Bit[1:0]: Reserved
P3:0x9C~ P3:0xDF	RSVD	—	—	Reserved
P3:0xE0	TP_E	0x00	RW	Bit[7:2]: Reserved Bit[1]: Select colorbar 0: Colorbar 1: Reserved Bit[0]: Enable colorbar
P3:0xE1	TP_DPC_AREA_SEL	0x0F	RW	Bit[7:4]: Reserved Bit[3]: area4_en Bit[2]: area3_en Bit[1]: area2_en Bit[0]: area1_en
P3:0xE2	TP_TEST_DPC_EN	0x00	RW	Bit[7:2]: Reserved Bit[1]: OTP DPC bad pixel pattern Bit[0]: Dynamic DPC bad pixel pattern
P3:0xE3	TP_OB1	0x40	RW	Bit[7:0]: tp_ob1
P3:0xE4	TP_OB2	0x20	RW	Bit[7:0]: tp_ob2
P3:0xE5	TP_OFFSET	0x20	RW	Bit[7:0]: tp_offset
P3:0xE6	TP_CB_PXWHITE	0xE0	RW	Bit[7:0]: tp_cb_pxwhite
P3:0xE7	TP_CB_PXBLACK	0x0A	RW	Bit[7:0]: tp_cb_pxblack
P3:0xE8	TP_DPC_PXWHITE	0xF0	RW	Bit[7:0]: tp_dpc_pxwhite
P3:0xE9	TP_DPC_PXBLACK	0x00	RW	Bit[7:0]: tp_dpc_pxblack

table 5-4 ISP registers (sheet 4 of 4)

address	register name	default value	R/W	description
P3:0xEA	TP_DPC_EDGE	0x8B	RW	Bit[7]: edge_en Bit[6:0]: edge_sel Select bad pixel pattern from DPC pattern to edge Range is [1:29]

5.5 dynamic DPC long frame [P4:0x00 ~ P4:0x22]

table 5-5 dynamic DPC long frame registers (sheet 1 of 3)

address	register name	default value	R/W	description
P4:0x00	DPC_CTRL00	0x1B	RW	Bit[7]: Enable tail Bit[6]: Enable general tail Bit[5]: Enable 33 cluster Bit[4]: Enable saturate cross cluster Bit[3]: Enable cross cluster Bit[2]: Manual mode Bit[1]: Black pixel Bit[0]: White pixel
P4:0x01	DPC_CTRL01	0x94	RW	Bit[7:6]: vnumlist_2 Bit[5:4]: vnumlist_1 Bit[3:2]: vnumlist_0 Bit[1]: Clipinterpg Bit[0]: Share buffer
P4:0x02	DPC_CTRL02	0x2E	RW	Bit[7:6]: Not used Bit[5:4]: Pixel order Bit[3:2]: Edge option Bit[1:0]: vnumlist_3
P4:0x03	DPC_CTRL03	0x24	RW	Bit[7]: Not used Bit[6:3]: wthreslist_0 Bit[2:0]: Maximum vnum
P4:0x04	DPC_CTRL04	0x12	RW	Bit[7:4]: wthreslist_2 Bit[3:0]: wthreslist_1
P4:0x05	DPC_CTRL05	0x41	RW	Bit[7:4]: bthresratio Bit[3:0]: wthreslist_3
P4:0x06	DPC_CTRL06	0x48	RW	Bit[7:4]: Status threshold Bit[3:0]: More connection case threshold
P4:0x07	DPC_CTRL07	0x84	RW	Bit[7:4]: Matching threshold Bit[3:0]: Status threshold step

table 5-5 dynamic DPC long frame registers (sheet 2 of 3)

address	register name	default value	R/W	description
P4:0x08	DPC_CTRL08	0x40	RW	Bit[7:4]: Adaptive pattern step Bit[3:0]: Adaptive pattern threshold
P4:0x09	DPC_CTRL09	0x00	RW	Bit[7:4]: Not used Bit[3:0]: Saturate
P4:0x0A	DPC_CTRL0A	0x03	RW	Bit[7:0]: gainlist_0[7:0]
P4:0x0B	DPC_CTRL0B	0x0F	RW	Bit[7:0]: gainlist_1[7:0]
P4:0x0C	DPC_CTRL0C	0x3F	RW	Bit[7:0]: gainlist_2[7:0]
P4:0x0D	DPC_CTRL0D	0x0F	RW	Bit[7:0]: dpclevellist_0
P4:0x0E	DPC_CTRL0E	0xFD	RW	Bit[7:0]: dpclevellist_1
P4:0x0F	DPC_CTRL0F	0xF5	RW	Bit[7:0]: dpclevellist_2
P4:0x10	DPC_CTRL10	0xF5	RW	Bit[7:0]: dpclevellist_3
P4:0x11	DPC_CTRL11	0x00	RW	Bit[7:3]: Not used Bit[2]: gainlist_0[8] Bit[1]: gainlist_1[8] Bit[0]: gainlist_2[8]
P4:0x12	DPC_CTRL12	0x00	RW	Bit[7]: Clkgatedisrec2 Bit[6]: Clkgatedisrec1 Bit[5]: Clkgatedisrec0 Bit[4]: Clkgatedissram Bit[3]: Clkgatedisbuf Bit[2]: Clkgatedissf Bit[1]: Clkgatedis Bit[0]: Manpxorder
P4:0x13	DPC_STAT13	–	R	Bit[7]: Not used Bit[6:0]: Bthre
P4:0x14	DPC_STAT14	–	R	Bit[7:5]: Not used Bit[4:0]: Wthre
P4:0x15	DPC_STAT15	–	R	Bit[7:5]: Not used Bit[4:0]: Thre1
P4:0x16	DPC_STAT16	–	R	Bit[7:0]: Thre2
P4:0x17	DPC_STAT17	–	R	Bit[7]: Not used Bit[6:0]: Thre3
P4:0x18	DPC_STAT18	–	R	Bit[7]: Not used Bit[6:0]: Thre4
P4:0x19	DPC_STAT19	–	R	Bit[7:4]: Level Bit[3:0]: Pconnected
P4:0x1A	DPC_STAT1A	–	R	Bit[7:3]: Not used Bit[2:0]: Vnum

table 5-5 dynamic DPC long frame registers (sheet 3 of 3)

address	register name	default value	R/W	description
P4:0x1B	DPC_EN	0x00	RW	Bit[7:1]: Not used Bit[0]: Enable DPC
P4:0x1C	DPC_CFG	0x22	RW	Bit[7:6]: Not used Bit[5]: auto_first_en Bit[4]: br_first Bit[3:2]: cfa_ptn Bit[1]: cen_global Bit[0]: sclk_gt_dis
P4:0x1D	DPC_HSIZE	0x07	RW	Bit[7:4]: Not used Bit[3:0]: hsize_dpc[11:8]
P4:0x1E	DPC_HSIZE	0x88	RW	Bit[7:0]: hsize_dpc[7:0]
P4:0x1F	DPC_MANU_HSIZE_EN	0x00	RW	Bit[7:1]: Not used Bit[0]: Enable manual hsize
P4:0x20	DPC_BIST_AUTO_MANUAL	0x00	RW	Bit[7:2]: Not used Bit[1]: bist_msel_en_dpc Bit[0]: bist_auto_en
P4:0x21	DPC_BIST_BYTE	0x00	RW	Bit[7:5]: Not used Bit[4:3]: bist_byte_depth Bit[2:0]: bist_byte_width
P4:0x22	DPC_BIST_INFO	—	R	Bit[7:2]: Not used Bit[1]: bist_error_flag Bit[0]: bist_test_done

5.6 dynamic DPC short frame [P4:0x80 - P4:0x9B]

table 5-6 dynamic DPC short frame registers (sheet 1 of 3)

address	register name	default value	R/W	description
P4:0x80	DPC_CTRL00	0x1B	RW	Bit[7]: Enable tail Bit[6]: Enable general tail Bit[5]: Enable 33 cluster Bit[4]: Enable saturate cross cluster Bit[3]: Enable cross cluster Bit[2]: Manual mode Bit[1]: Black pixel Bit[0]: White pixel

table 5-6 dynamic DPC short frame registers (sheet 2 of 3)

address	register name	default value	R/W	description
P4:0x81	DPC_CTRL01	0x94	RW	Bit[7:6]: vnumlist_2 Bit[5:4]: vnumlist_1 Bit[3:2]: vnumlist_0 Bit[1]: Clipinterpg Bit[0]: Share buffer
P4:0x82	DPC_CTRL02	0x2E	RW	Bit[7:6]: Not used Bit[5:4]: Pixel order Bit[3:2]: Edge option Bit[1:0]: vnumlist_3
P4:0x83	DPC_CTRL03	0x24	RW	Bit[7]: Not used Bit[6:3]: wthreslist_0 Bit[2:0]: Maxvnum
P4:0x84	DPC_CTRL04	0x12	RW	Bit[7:4]: wthreslist_2 Bit[3:0]: wthreslist_1
P4:0x85	DPC_CTRL05	0x41	RW	Bit[7:4]: bthresratio Bit[3:0]: wthreslist_3
P4:0x86	DPC_CTRL06	0x48	RW	Bit[7:4]: Statusthre Bit[3:0]: More connection case threshold
P4:0x87	DPC_CTRL07	0x84	RW	Bit[7:4]: Matching threshold Bit[3:0]: Status threshold step
P4:0x88	DPC_CTRL08	0x40	RW	Bit[7:4]: Adaptive pattern step Bit[3:0]: Adaptive pattern threshold
P4:0x89	DPC_CTRL09	0x00	RW	Bit[7:4]: Not used Bit[3:0]: Saturate
P4:0x8A	DPC_CTRL0A	0x03	RW	Bit[7:0]: gainlist_0[7:0]
P4:0x8B	DPC_CTRL0B	0x0F	RW	Bit[7:0]: gainlist_1[7:0]
P4:0x8C	DPC_CTRL0C	0x3F	RW	Bit[7:0]: gainlist_2[7:0]
P4:0x8D	DPC_CTRL0D	0x0F	RW	Bit[7:0]: dpclevellist_0
P4:0x8E	DPC_CTRL0E	0xFD	RW	Bit[7:0]: dpclevellist_1
P4:0x8F	DPC_CTRL0F	0xF5	RW	Bit[7:0]: dpclevellist_2
P4:0x90	DPC_CTRL10	0xF5	RW	Bit[7:0]: dpclevellist_3
P4:0x91	DPC_CTRL11	0x00	RW	Bit[7:3]: Not used Bit[2]: gainlist_0[8] Bit[1]: gainlist_1[8] Bit[0]: gainlist_2[8]

table 5-6 dynamic DPC short frame registers (sheet 3 of 3)

address	register name	default value	R/W	description	
P4:0x92	DPC_CTRL12	0x00	RW	Bit[7]:	Clkgatedisrec2
				Bit[6]:	Clkgatedisrec1
				Bit[5]:	Clkgatedisrec0
				Bit[4]:	Clkgatedissram
				Bit[3]:	Clkgatedisbuf
				Bit[2]:	Clkgatedissf
				Bit[1]:	Clkgatedis
				Bit[0]:	Manpxorder
P4:0x93	DPC_STAT13	–	R	Bit[7]:	Not used
				Bit[6:0]:	Bthre
P4:0x94	DPC_STAT14	–	R	Bit[7:5]:	Not used
				Bit[4:0]:	Wthre
P4:0x95	DPC_STAT15	–	R	Bit[7:5]:	Not used
				Bit[4:0]:	Thre1
P4:0x96	DPC_STAT16	–	R	Bit[7:0]:	Thre2
P4:0x97	DPC_STAT17	–	R	Bit[7]:	Not used
				Bit[6:0]:	Thre3
P4:0x98	DPC_STAT18	–	R	Bit[7]:	Not used
				Bit[6:0]:	Thre4
P4:0x99	DPC_STAT19	–	R	Bit[7:4]:	Level
				Bit[3:0]:	Pconnected
P4:0x9A	DPC_STAT1A	–	R	Bit[7:3]:	Not used
				Bit[2:0]:	Vnum
P4:0x9B	DPC_EN	0x00	RW	Bit[7:1]:	Not used
				Bit[0]:	Enable DPC

5.7 BLC [P5:0x00 ~ P5:0xDB]

table 5-7 BLC registers (sheet 1 of 7)

address	register name	default value	R/W	description
P5:0x00	REG_BLC_00	0xF7	RW	Bit[7]: r_blc_en_o Bit[6]: Not used Bit[5]: r_gain_trig_priority_o Bit[4]: r_rst_trig_opt_o Bit[3]: r_target_adj_dis_o Bit[2]: r_off_cmp_en_o Bit[1]: r_dither_en_o Bit[0]: r_mf_en_o
P5:0x01	REG_BLC_01	0x60	RW	Bit[7]: r_dcblc_dis_reg_en_o Bit[6]: r_dcblc_en_o Bit[5]: r_kcoef_man_en_o Bit[4]: r_off_man_en_o Bit[3]: r_zero_in_out_en_o Bit[2]: r_blk_in_out_en_o Bit[1:0]: r_byp_mode_o
P5:0x02	REG_BLC_02	0x01	RW	Bit[7:0]: r_blk_lvl_target_o[15:8]
P5:0x03	REG_BLC_03	0x00	RW	Bit[7:0]: r_blk_lvl_target_o[7:0]
P5:0x04	REG_BLC_04	0x00	RW	Bit[7:4]: Not used Bit[3:0]: r_hwin_off_o[11:8]
P5:0x05	REG_BLC_05	0x00	RW	Bit[7:0]: r_hwin_off_o[7:0]
P5:0x06	REG_BLC_06	0x00	RW	Bit[7:4]: Not used Bit[3:0]: r_hwin_pad_o[11:8]
P5:0x07	REG_BLC_07	0x04	RW	Bit[7:0]: r_hwin_pad_o[7:0]
P5:0x08	REG_BLC_08	0x04	RW	Bit[7:5]: Not used Bit[4:0]: r_bl_start_o[4:0]
P5:0x09	REG_BLC_09	0x0B	RW	Bit[7:5]: Not used Bit[4:0]: r_bl_end_o[4:0]
P5:0x0A	REG_BLC_0A	0x02	RW	Bit[7:3]: Not used Bit[2:0]: r_off_lim_th_o[10:8]
P5:0x0B	REG_BLC_0B	0x00	RW	Bit[7:0]: r_off_lim_th_o[7:0]
P5:0x0C~ P5:0x0D	RSVD	—	—	Reserved
P5:0x0E	REG_BLC_0E	0x00	RW	Bit[7:5]: Not used Bit[4:0]: r_mf_th_o[4:0]

table 5-7 BLC registers (sheet 2 of 7)

address	register name	default value	R/W	description
P5:0x0F	REG_BLC_0F	0x82	RW	Bit[7]: r_exp_chg_trig_en_o Bit[6]: r_set_zb_o Bit[5:4]: Not used Bit[3]: r_v14_one_channel_o Bit[2:0]: Not used
P5:0x10	REG_BLC_10	0xF0	RW	Bit[7]: r_off_trig_en_o Bit[6]: r_gain_chg_trig_en_o Bit[5]: r_fmt_chg_trig_en_o Bit[4]: r_rst_trig_en_o Bit[3]: r_man_avg_en_o Bit[2]: r_man_trig_o Bit[1]: r_off_frz_en_o Bit[0]: r_off_always_up_o
P5:0x11	REG_BLC_11	0xFF	RW	Bit[7]: r_off_cmp_man_en_o Bit[6]: r_off_chg_mf_en_o Bit[5]: r_fmt_chg_mf_en_o Bit[4]: r_gain_chg_mf_en_o Bit[3]: r_rst_mf_mode_o Bit[2]: r_off_chg_mf_mode_o Bit[1]: r_fmt_chg_mf_mode_o Bit[0]: r_gain_chg_mf_mode_o
P5:0x12	REG_BLC_12	0x08	RW	Bit[7:0]: r_rst_trig_fn_o
P5:0x13	REG_BLC_13	0x02	RW	Bit[7:0]: r_fmt_trig_fn_o
P5:0x14	REG_BLC_14	0x02	RW	Bit[7:0]: r_gain_trig_fn_o
P5:0x15	REG_BLC_15	0x02	RW	Bit[7:0]: r_off_trig_fn_o
P5:0x16	REG_BLC_16	0x00	RW	Bit[7:2]: Not used Bit[1:0]: r_off_trig_th_o[9:8]
P5:0x17	REG_BLC_17	0x04	RW	Bit[7:0]: r_off_trig_th_o[7:0]
P5:0x18~ P5:0x1F	RSVD	–	–	Reserved
P5:0x20	REG_BLC_20	0x00	RW	Bit[7:0]: r_off_cmp_th000_o
P5:0x21	REG_BLC_21	0x00	RW	Bit[7:0]: r_off_cmp_k000_o
P5:0x22	REG_BLC_22	0x00	RW	Bit[7:0]: r_off_cmp_th001_o
P5:0x23	REG_BLC_23	0x00	RW	Bit[7:0]: r_off_cmp_k001_o
P5:0x24	REG_BLC_24	0x00	RW	Bit[7:0]: r_off_cmp_th010_o
P5:0x25	REG_BLC_25	0x00	RW	Bit[7:0]: r_off_cmp_k010_o
P5:0x26	REG_BLC_26	0x00	RW	Bit[7:0]: r_off_cmp_th011_o
P5:0x27	REG_BLC_27	0x00	RW	Bit[7:0]: r_off_cmp_k011_o

table 5-7 BLC registers (sheet 3 of 7)

address	register name	default value	R/W	description
P5:0x28~ P5:0x2F	RSVD	—	—	Reserved
P5:0x30	REG_BLC_30	0x00	RW	Bit[7:3]: Not used Bit[2:0]: r_off_man000_o[10:8]
P5:0x31	REG_BLC_31	0x00	RW	Bit[7:0]: r_off_man000_o[7:0]
P5:0x32	REG_BLC_32	0x00	RW	Bit[7:3]: Not used Bit[2:0]: r_off_man001_o[10:8]
P5:0x33	REG_BLC_33	0x00	RW	Bit[7:0]: r_off_man001_o[7:0]
P5:0x34	REG_BLC_34	0x00	RW	Bit[7:3]: Not used Bit[2:0]: r_off_man010_o[10:8]
P5:0x35	REG_BLC_35	0x00	RW	Bit[7:0]: r_off_man010_o[7:0]
P5:0x36	REG_BLC_36	0x00	RW	Bit[7:3]: Not used Bit[2:0]: r_off_man011_o[10:8]
P5:0x37	REG_BLC_37	0x00	RW	Bit[7:0]: r_off_man011_o[7:0]
P5:0x38	REG_BLC_38	0x00	RW	Bit[7:3]: Not used Bit[2:0]: r_off_man100_o[10:8]
P5:0x39	REG_BLC_39	0x00	RW	Bit[7:0]: r_off_man100_o[7:0]
P5:0x3A	REG_BLC_3A	0x00	RW	Bit[7:3]: Not used Bit[2:0]: r_off_man101_o[10:8]
P5:0x3B	REG_BLC_3B	0x00	RW	Bit[7:0]: r_off_man101_o[7:0]
P5:0x3C	REG_BLC_3C	0x00	RW	Bit[7:3]: Not used Bit[2:0]: r_off_man110_o[10:8]
P5:0x3D	REG_BLC_3D	0x00	RW	Bit[7:0]: r_off_man110_o[7:0]
P5:0x3E	REG_BLC_3E	0x00	RW	Bit[7:3]: Not used Bit[2:0]: r_off_man111_o[10:8]
P5:0x3F	REG_BLC_3F	0x00	RW	Bit[7:0]: r_off_man111_o[7:0]
P5:0x40	REG_BLC_40	0x00	RW	Bit[7:6]: Not used Bit[5:0]: r_zl_start_o
P5:0x41	REG_BLC_41	0x0F	RW	Bit[7:6]: Not used Bit[5:0]: r_zl_end_o
P5:0x42	REG_BLC_42	0x01	RW	Bit[7:2]: Not used Bit[1:0]: r_kcoef_b_man_o[9:8]
P5:0x43	REG_BLC_43	0x00	RW	Bit[7:0]: r_kcoef_b_man_o[7:0]
P5:0x44	REG_BLC_44	0x01	RW	Bit[7:2]: Not used Bit[1:0]: r_kcoef_gb_man_o[9:8]

table 5-7 BLC registers (sheet 4 of 7)

address	register name	default value	R/W	description
P5:0x45	REG_BLC_45	0x00	RW	Bit[7:0]: r_kcoef_gb_man_o[7:0]
P5:0x46	REG_BLC_46	0x01	RW	Bit[7:2]: Not used Bit[1:0]: r_kcoef_gr_man_o[9:8]
P5:0x47	REG_BLC_47	0x00	RW	Bit[7:0]: r_kcoef_gr_man_o[7:0]
P5:0x48	REG_BLC_48	0x01	RW	Bit[7:2]: Not used Bit[1:0]: r_kcoef_r_man_o[9:8]
P5:0x49	REG_BLC_49	0x00	RW	Bit[7:0]: r_kcoef_r_man_o[7:0]
P0:0x4A~ P0:0x4B	RSVD	–	–	Reserved
P5:0x4C	REG_BLC_4C	0x3F	RW	Bit[7:6]: Not used Bit[5:0]: r_off_avg_weight_o
P5:0x4D	REG_BLC_4D	0x00	RW	Bit[7:1]: Not used Bit[0]: r_rnd_gain_th_o[8]
P5:0x4E	REG_BLC_4E	0x00	RW	Bit[7:0]: r_rnd_gain_th_o[7:0]
P5:0x4F~ P5:0x5F	RSVD	–	–	Reserved
P5:0x60	REG_BLC_60	–	R	Bit[7:0]: blc_offset000[15:8]
P5:0x61	REG_BLC_61	–	R	Bit[7:0]: blc_offset000[7:0]
P5:0x62	REG_BLC_62	–	R	Bit[7:0]: blc_offset001[15:8]
P5:0x63	REG_BLC_63	–	R	Bit[7:0]: blc_offset001[7:0]
P5:0x64	REG_BLC_64	–	R	Bit[7:0]: blc_offset010[15:8]
P5:0x65	REG_BLC_65	–	R	Bit[7:0]: blc_offset010[7:0]
P5:0x66	REG_BLC_66	–	R	Bit[7:0]: blc_offset011[15:8]
P5:0x67	REG_BLC_67	–	R	Bit[7:0]: blc_offset011[7:0]
P5:0x68	REG_BLC_68	–	R	Bit[7:0]: blc_offset100[15:8]
P5:0x69	REG_BLC_69	–	R	Bit[7:0]: blc_offset100[7:0]
P5:0x6A	REG_BLC_6A	–	R	Bit[7:0]: blc_offset101[15:8]
P5:0x6B	REG_BLC_6B	–	R	Bit[7:0]: blc_offset101[7:0]
P5:0x6C	REG_BLC_6C	–	R	Bit[7:0]: blc_offset110[15:8]
P5:0x6D	REG_BLC_6D	–	R	Bit[7:0]: blc_offset110[7:0]
P5:0x6E	REG_BLC_6E	–	R	Bit[7:0]: blc_offset111[15:8]
P5:0x6F	REG_BLC_6F	–	R	Bit[7:0]: blc_offset111[7:0]

table 5-7 BLC registers (sheet 5 of 7)

address	register name	default value	R/W	description
P5:0x70	REG_BLC_70	–	R	Bit[7:0]: off_cmp_k_00
P5:0x71	REG_BLC_71	–	R	Bit[7:0]: off_cmp_k_01
P5:0x72	REG_BLC_72	–	R	Bit[7:0]: off_cmp_k_10
P5:0x73	REG_BLC_73	–	R	Bit[7:0]: off_cmp_k_11
P5:0x74	RSVD	–	–	Reserved
P5:0x75	REG_BLC_75	–	R	Bit[7:0]: ana_real_gain_l[7:0]
P5:0x76	RSVD	–	–	Reserved
P5:0x77	REG_BLC_77	–	R	Bit[7:0]: ana_real_gain_s[7:0]
P5:0x78~ P5:0x7F	RSVD	–	–	Reserved
P5:0x80	REG_BLC_80	–	R	Bit[7:3]: Not used Bit[2:0]: cmp_00_l[10:8]
P5:0x81	REG_BLC_81	–	R	Bit[7:0]: cmp_00_l[7:0]
P5:0x82	REG_BLC_82	–	R	Bit[7:3]: Not used Bit[2:0]: cmp_01_l[10:8]
P5:0x83	REG_BLC_83	–	R	Bit[7:0]: cmp_01_l[7:0]
P5:0x84	REG_BLC_84	–	R	Bit[7:3]: Not used Bit[2:0]: cmp_10_l[10:8]
P5:0x85	REG_BLC_85	–	R	Bit[7:0]: cmp_10_l[7:0]
P5:0x86	REG_BLC_86	–	R	Bit[7:3]: Not used Bit[2:0]: cmp_11_l[10:8]
P5:0x87	REG_BLC_87	–	R	Bit[7:0]: cmp_11_l[7:0]
P5:0x88	REG_BLC_88	–	R	Bit[7:2]: Not used Bit[1:0]: kcoef_b[9:8]
P5:0x89	REG_BLC_89	–	R	Bit[7:0]: kcoef_b[7:0]
P5:0x8A	REG_BLC_8A	–	R	Bit[7:2]: Not used Bit[1:0]: kcoef_gb[9:8]
P5:0x8B	REG_BLC_8B	–	R	Bit[7:0]: kcoef_gb[7:0]
P5:0x8C	REG_BLC_8C	–	R	Bit[7:2]: Not used Bit[1:0]: kcoef_gr[9:8]
P5:0x8D	REG_BLC_8D	–	R	Bit[7:0]: kcoef_gr[7:0]
P5:0x8E	REG_BLC_8E	–	R	Bit[7:2]: Not used Bit[1:0]: kcoef_r[9:8]

table 5-7 BLC registers (sheet 6 of 7)

address	register name	default value	R/W	description	
P5:0x8F	REG_BLC_8F	–	R	Bit[7:0]:	kcoef_r[7:0]
P5:0x90	REG_BLC_90	0x00	RW	Bit[7]: Bit[6]: Bit[5]: Bit[4:3]: Bit[2]: Bit[1]: Bit[0]:	cmp_flip_auto_old_en kcoef_flip_auto_old_en off_read_prec_sel Not used hdr_mode_i r_format_trig_beh r_gain_trig_beh
P5:0x91	REG_BLC_91	0x00	RW	Bit[7]: Bit[6]: Bit[5]: Bit[4]: Bit[3]: Bit[2]: Bit[1]: Bit[0]:	cmp_mirror_man cmp_mirror_man_en cmp_flip_man cmp_flip_man_en kcoef_mirror_man kcoef_mirror_man_en kcoef_flip_man kcoef_flip_man_en
P5:0x92	REG_BLC_92	0x00	RW	Bit[7:2]: Bit[1]: Bit[0]:	Not used off_man_mirror_man_en off_man_mirror_man
P5:0x93~ P5:0x9F	RSVD	–	–	Reserved	
P5:0xA0	REG_BLC_A0	0x00	RW	Bit[7:3]: Bit[2:0]:	Not used cmp_00_s_i[10:8]
P5:0xA1	REG_BLC_A1	0x00	RW	Bit[7:0]:	cmp_00_s_i[7:0]
P5:0xA2	REG_BLC_A2	0x00	RW	Bit[7:3]: Bit[2:0]:	Not used cmp_01_s_i[10:8]
P5:0xA3	REG_BLC_A3	0x00	RW	Bit[7:0]:	cmp_01_s_i[7:0]
P5:0xA4	REG_BLC_A4	0x00	RW	Bit[7:3]: Bit[2:0]:	Not used cmp_10_s_i[10:8]
P5:0xA5	REG_BLC_A5	0x00	RW	Bit[7:0]:	cmp_10_s_i[7:0]
P5:0xA6	REG_BLC_A6	0x00	RW	Bit[7:3]: Bit[2:0]:	Not used cmp_00_s_i[10:8]
P5:0xA7	REG_BLC_A7	0x00	RW	Bit[7:0]:	cmp_00_s_i[7:0]
P5:0xA8~ P5:0xAF	RSVD	–	–	Reserved	
P5:0xB0	REG_BLC_B0	0x00	RW	Bit[7:1]: Bit[0]:	Not used r_thres_en
P5:0xB1~ P5:0xB2	RSVD	–	–	Reserved	

table 5-7 BLC registers (sheet 7 of 7)

address	register name	default value	R/W	description
P5:0xB3	REG_BLC_B3	0x00	RW	Bit[7:3]: Not used Bit[2:0]: hist_thres_reg[10:8]
P5:0xB4	REG_BLC_B4	0x00	RW	Bit[7:0]: hist_thres_reg[7:0]
P5:0xB5~ P5:0xB6	RSVD	–	–	Reserved
P5:0xB7	REG_BLC_B7	0x00	RW	Bit[7:3]: Not used Bit[2:0]: hist_thres_s_reg[10:8]
P5:0xB8	REG_BLC_B8	0x00	RW	Bit[7:0]: hist_thres_s_reg[7:0]
P5:0xB9	RSVD	–	–	Reserved
P5:0xBA	REG_BLC_BA	0x00	RW	Bit[7:3]: Not used Bit[2]: r_gate_data_en Bit[1]: r_gate_clk_en Bit[0]: r_dummy_out_en
P5:0xBB~ P5:0xBC	RSVD	–	–	Reserved
P5:0xBD	REG_BLC_BD	0x97	RW	Bit[7:0]: Row center[7:0]
P5:0xBE	REG_BLC_BE	0xD2	RW	Bit[7:0]: Column center[7:0]
P5:0xBF	REG_BLC_BF	0x00	RW	Bit[7:0]: Row offset[7:0]
P5:0xC0	REG_BLC_C0	0x00	RW	Bit[7:0]: Column offset[7:0]
P5:0xC1	REG_BLC_C1	0x30	RW	Bit[7]: Not used Bit[6:5]: Row center[9:8] Bit[4:3]: Column center[9:8] Bit[2:0]: Row offset[10:8]
P5:0xC2	REG_BLC_C2	0x00	RW	Bit[7:4]: Not used Bit[3:0]: Column offset[11:8]
P5:0xC3~ P5:0xDA	RSVD	–	–	Reserved
P5:0xDB	REG_BLC_DB	0x00	RW	Bit[7:4]: Not used Bit[3]: offset_reg_en Bit[2]: kscg_otp_en Bit[1]: scg_en Bit[0]: scg_sel

5.8 OTP DPC [P6:0x04 ~ P6:0x2B]

table 5-8 OTP DPC registers (sheet 1 of 3)

address	register name	default value	R/W	description
P6:0x04	CTRL_04	0x00	RW	Bit[7]: rgbc_en Bit[6]: bin_cluster_sel Bit[5]: rgbc_bin_mir_sw Bit[4]: man_inc_en Bit[3]: disable_mf Bit[2]: disable_offset Bit[1]: mirror_opt Bit[0]: disable_bin
P6:0x05	CTRL_05	0x68	RW	Bit[7]: Not used Bit[6]: rgbc_c_recover_sel Bit[5]: recov_method Bit[4]: fixed_replace Bit[3]: fixed_ptn Bit[2]: flip_opt Bit[1]: expo_en Bit[0]: gain_en
P6:0x06	CTRL_06	0x00	RW	Bit[7]: Not used Bit[6:0]: expo_constrain[14:8]
P6:0x07	CTRL_07	0x00	RW	Bit[7:0]: expo_constrain[7:0]
P6:0x08	CTRL_08	0x07	RW	Bit[7:6]: Not used Bit[5:0]: gain_constrain[5:0]
P6:0x09	CTRL_09	0x48	RW	Bit[7]: xy_end_sel Bit[6]: vsync_rst_en Bit[5]: rgbc_bin_opt Bit[4]: thre_en Bit[3:0]: thre_l
P6:0x0A	CTRL_0A	0x01	RW	Bit[7:5]: Not used Bit[4:0]: man_x_even_inc
P6:0x0B	CTRL_0B	0x01	RW	Bit[7:5]: Not used Bit[4:0]: man_x_odd_inc
P6:0x0C	CTRL_0C	0x01	RW	Bit[7:5]: Not used Bit[4:0]: man_y_even_inc
P6:0x0D	CTRL_0D	0x01	RW	Bit[7:5]: Not used Bit[4:0]: man_y_odd_inc
P6:0x0E~ P6:0x0F	RSVD	–	–	Reserved
P6:0x10	CTRL_10	–	R	Bit[7:4]: Not used Bit[3:0]: x_offset[11:8]

table 5-8 OTP DPC registers (sheet 2 of 3)

address	register name	default value	R/W	description
P6:0x11	CTRL_11	–	R	Bit[7:0]: x_offset[7:0]
P6:0x12	CTRL_12	–	R	Bit[7:3]: Not used Bit[2:0]: y_offset[10:8]
P6:0x13	CTRL_13	–	R	Bit[7:0]: y_offset[7:0]
P6:0x14	CTRL_14	–	R	Bit[7:5]: Not used Bit[4:0]: x_even_inc
P6:0x15	CTRL_15	–	R	Bit[7:5]: Not used Bit[4:0]: x_odd_inc
P6:0x16	CTRL_16	–	R	Bit[7:5]: Not used Bit[4:0]: y_even_inc
P6:0x17	CTRL_17	–	R	Bit[7:5]: Not used Bit[4:0]: y_odd_inc
P6:0x18~ P6:0x1F	RSVD	–	–	Reserved
P6:0x20	CTRL_20	0x00	RW	Bit[7:5]: Not used Bit[4:0]: man_x_offset[12:8]
P6:0x21	CTRL_21	0x00	RW	Bit[7:0]: man_x_offset[7:0]
P6:0x22	CTRL_22	0x00	RW	Bit[7:4]: Not used Bit[3:0]: man_y_offset[11:8]
P6:0x23	CTRL_23	0x00	RW	Bit[7:0]: man_y_offset[7:0]
P6:0x24	RSVD	–	–	Reserved
P6:0x25	CTRL_25	0x00	RW	Bit[7:0]: end_addr_bin
P6:0x26	CTRL_26	0x00	RW	Bit[7]: y_bin_man_en Bit[6]: y_bin4_man Bit[5]: y_bin3_man Bit[4]: y_bin2_man Bit[3]: x_bin_man_en Bit[2]: x_bin4_man Bit[1]: x_bin3_man Bit[0]: x_bin2_man
P6:0x27	RSVD	–	–	Reserved
P6:0x28	CTRL_28	0x00	RW	Bit[7]: Not used Bit[6:0]: expo_constrain[14:8]
P6:0x29	CTRL_29	0x00	RW	Bit[7:0]: expo_constrain[7:0]
P6:0x2A	CTRL_2A	0x07	RW	Bit[7:6]: Not used Bit[5:0]: gain_constrain

table 5-8 OTP DPC registers (sheet 3 of 3)

address	register name	default value	R/W	description
P6:0x2B	CTRL_2B	0x08	RW	Bit[7:4]: Not used Bit[3:0]: thre_s

5.9 DCG combine [P7:0x00 - P7:0x38]

table 5-9 DCG combine registers (sheet 1 of 3)

address	register name	default value	R/W	description
P7:0x00	CTRL_00	0x19	RW	Bit[7:6]: Not used Bit[5]: Enable dither Bit[4:3]: Not used Bit[2]: offset_only_en Bit[1]: ratio_only_en Bit[0]: Reserved
P7:0x01	CTRL_01	0x03	RW	Bit[7:6]: Not used Bit[5:4]: comb_opt Bit[3:2]: Not used Bit[1]: expo_update Bit[0]: Enable gain update
P7:0x02	CTRL_02	0x60	RW	Bit[7:0]: comb_tl
P7:0x03	CTRL_03	0xF0	RW	Bit[7:0]: comb_th
P7:0x04	CTRL_04	0x04	RW	Bit[7:0]: blc_target[15:8]
P7:0x05	CTRL_05	0x00	RW	Bit[7:0]: blc_target[7:0]
P7:0x06	CTRL_06	0x02	RW	Bit[7:6]: Not used Bit[5:0]: cur_gain[13:8]
P7:0x07	CTRL_07	0x00	RW	Bit[7:0]: cur_gain[7:0]
P7:0x08~ P7:0x0A	RSVD	–	–	Reserved
P7:0x0B	CTRL_0B	0x10	RW	Bit[7:6]: Not used Bit[5:0]: ratio_range
P7:0x0C	CTRL_0C	0x04	RW	Bit[7:4]: Not used Bit[3:0]: offset_range[11:8]
P7:0x0D	CTRL_0D	0x00	RW	Bit[7:0]: offset_range[7:0]
P7:0x0E~ P7:0x13	RSVD	–	–	Reserved

table 5-9 DCG combine registers (sheet 2 of 3)

address	register name	default value	R/W	description
P7:0x14	CTRL_14	0x00	RW	Bit[7:5]: Not used Bit[4:0]: offset_table[12:8]
P7:0x15	CTRL_15	0x00	RW	Bit[7:0]: offset_table[7:0]
P7:0x16~ P7:0x17	RSVD	–	–	Reserved
P7:0x18	STAT_18	–	R	Bit[7:6]: Not used Bit[5:0]: gain_cur_calc_i[13:8]
P7:0x19	STAT_19	–	R	Bit[7:0]: gain_cur_calc_i[7:0]
P7:0x1A	STAT_1A	–	R	Bit[7:0]: cur_offset_calc_i[15:8]
P7:0x1B	STAT_1B	–	R	Bit[7:0]: cur_offset_calc_i[7:0]
P7:0x1C	STAT_1C	–	R	Bit[7:2]: Not used Bit[1:0]: cur_ratio_calc_i[17:16]
P7:0x1D	STAT_1D	–	R	Bit[7:0]: cur_ratio_calc_i[15:8]
P7:0x1E	STAT_1E	–	R	Bit[7:0]: cur_ratio_calc_i[7:0]
P7:0x1F	STAT_1F	–	R	Bit[7:2]: Not used Bit[1:0]: d_gain_ratio_i[17:16]
P7:0x20	STAT_20	–	R	Bit[7:0]: d_gain_ratio_i[15:8]
P7:0x21	STAT_21	–	R	Bit[7:0]: d_gain_ratio_i[7:0]
P7:0x22	STAT_22	–	R	Bit[7:0]: min_ratio_i[23:16]
P7:0x23	STAT_23	–	R	Bit[7:0]: min_ratio_i[15:8]
P7:0x24	STAT_24	–	R	Bit[7:0]: min_ratio_i[7:0]
P7:0x25	STAT_25	–	R	Bit[7:1]: Not used Bit[0]: max_ratio_i[24:24]
P7:0x26	STAT_26	–	R	Bit[7:0]: max_ratio_i[23:16]
P7:0x27	STAT_27	–	R	Bit[7:0]: max_ratio_i[15:8]
P7:0x28	STAT_28	–	R	Bit[7:0]: max_ratio_i[7:0]
P7:0x29	STAT_29	–	R	Bit[7:1]: Reserved Bit[0]: sta_eof_i
P7:0x2A	STAT_2A	–	R	Bit[7:1]: Reserved Bit[0]: sta_stable_flag_i
P7:0x2B	STAT_2B	–	R	Bit[7:0]: cur_offset_sta_i[15:8]
P7:0x2C	STAT_2C	–	R	Bit[7:0]: cur_offset_sta_i[7:0]

table 5-9 DCG combine registers (sheet 3 of 3)

address	register name	default value	R/W	description
P7:0x2D	STAT_2D	–	R	Bit[7:1]: Reserved Bit[0]: cur_ratio_sta_i[24]
P7:0x2E	STAT_2E	–	R	Bit[7:0]: cur_ratio_sta_i[23:16]
P7:0x2F	STAT_2F	–	R	Bit[7:6]: Reserved Bit[5:0]: gain_cur_calc_i[13:8]
P7:0x30	CTRL_30	–	R	Bit[7:0]: cur_ratio_sta_i[15:8]
P7:0x31	CTRL_31	–	R	Bit[7:0]: cur_ratio_sta_i[7:0]
P7:0x32	CTRL_32	–	R	Bit[7:0]: version_i[15:8]
P7:0x33	CTRL_33	–	R	Bit[7:0]: version_i[7:0]
P7:0x34	CTRL_34	0x00	RW	Bit[7:6]: Reserved Bit[5]: dcg_en Bit[4]: bist_msel_en Bit[3]: bist_en Bit[2]: sclk_gt_dis Bit[1]: cen_global Bit[0]: mod_cen
P7:0x35	CTRL_35	0x00	RW	Bit[7:5]: Reserved Bit[4]: compress_en Bit[3:0]: bist_byte[3:0]
P7:0x36	CTRL_36	0x80	RW	Bit[7:0]: kscg_target
P7:0x37	CTRL_37	0x80	RW	Bit[7:4]: Reserved Bit[3]: autobist_en Bit[2:0]: bist_byte[6:4]
P7:0x38	CTRL_38	–	R	Bit[7:2]: Reserved Bit[1]: test_done_644x48 Bit[0]: error_flag_644x48

5.10 OTP [P9:0x80 - P9:0xC3]

table 5-10 OTP registers (sheet 1 of 4)

address	register name	default value	R/W	description
P9:0x80	CTRL_80	0x00	RW	Bit[7]: otp_pgenb (read only) Bit[6:1]: Reserved Bit[0]: otp_pgm (write only)

table 5-10 OTP registers (sheet 2 of 4)

address	register name	default value	R/W	description
P9:0x81	CTRL_81	0x00	RW	Bit[7]: otp_load_o load ongoing (read only) Bit[6]: Reserved Bit[5]: otp_bist_err (read only) Bit[4]: otp_bist_done (read only) Bit[3]: Reserved Bit[2]: bist_result_clr (write only) Bit[1]: autoload_m (write only) Bit[0]: otp_rd
P9:0x82	CTRL_82	0x2E	RW	Bit[7:0]: pgm_ctrl Control program strobe pulse
P9:0x83	CTRL_83	0x0E	RW	Bit[7:0]: rd_ctrl Control load strobe pulse
P9:0x84	CTRL_84	0x40	RW	Bit[7]: Disable program Bit[6]: Manual mode select 0: Auto 1: Manual Bit[5:1]: Reserved Bit[0]: bank_sram_switch
P9:0x85	CTRL_85	0x1B	RW	Bit[7]: Not used Bit[6]: otp_bist_comp_val Bit[5]: otp_bist_sel Bit[4]: otp_bist_enable Bit[3]: otp_pwup_load Bit[2]: otp_wkup_load Bit[1:0]: Not used
P9:0x86	RSVD	–	–	Reserved
P9:0x87	CTRL_87	0x36	RW	Bit[7:4]: Not used Bit[3:0]: PS2CS[9:6] PS2CS[5:0] is always zero
P9:0x88	CTRL_88	0x00	RW	Bit[7:0]: adr_stt_pt[15:8]
P9:0x89	CTRL_89	0x00	RW	Bit[7:0]: adr_stt_pt[7:0]
P9:0x8A	CTRL_8A	0x00	RW	Bit[7:0]: adr_end_pt[15:8]
P9:0x8B	CTRL_8B	0xFF	RW	Bit[7:0]: adr_end_pt[7:0]
P9:0x8C	CTRL_8C	0x00	RW	Bit[7:1]: Not used Bit[0]: otp_disable
P9:0x8D	CTRL_8D	0x00	RW	Bit[7:0]: otp_rst
P9:0x8E	CTRL_8E	–	R	Bit[7:1]: Not used Bit[0]: otp_bist_err_adr[8]
P9:0x8F	CTRL_8F	–	R	Bit[7:0]: otp_bist_err_adr[7:0]

table 5-10 OTP registers (sheet 3 of 4)

address	register name	default value	R/W	description
P9:0x90	CTRL_90	0x00	RW	Bit[7:2]: Not used Bit[1:0]: otp_adr_base[9:8]
P9:0x91	CTRL_91	0x00	RW	Bit[7:0]: otp_adr_base[7:0]
P9:0x92	CTRL_92	0x16	RW	Bit[7:0]: t_pgenb
P9:0x93	CTRL_93	0x66	RW	Bit[7:0]: t_vddq
P9:0x94	CTRL_94	0x12	RW	Bit[7:0]: strobe_gap_pgm
P9:0x95	CTRL_95	0x03	RW	Bit[7:6]: Not used Bit[5:0]: strobe_gap_load
P9:0x96	CTRL_96	0x1F	RW	Bit[7:0]: cnt_blen
P9:0x97	CTRL_97	0x32	RW	Bit[7:0]: t_aen_sup[10:3]
P9:0x98	CTRL_98	0x31	RW	Bit[7:4]: Not used Bit[3:0]: For control rc_s_rdy
P9:0x99	CTRL_99	0x10	RW	Bit[7:0]: t_load_sup[8:1]
P9:0x9A	CTRL_9A	0x00	RW	Bit[7]: Not used Bit[6]: otp_cis_sram_sel Bit[5]: mem_addr_sel Bit[4:0]: otp_sram_h5
P9:0x9B	RSVD	–	–	Reserved
P9:0x9C	CTRL_9C	0x51	RW	Bit[7:0]: For control frq_hd
P9:0x9D	CTRL_9D	0x00	RW	Bit[7]: otp_int_addr_fix_en Bit[6:5]: Not used Bit[4]: otp_byt_rd_en Bit[3]: win_empty_en Bit[2]: win_nvalid_en Bit[1]: win_mem_offset_en Bit[0]: dp_isp_en
P9:0x9E	CTRL_9E	0x20	RW	Bit[7:0]: fifo_depth
P9:0x9F	CTRL_9F	0x00	RW	Bit[7]: WEEN Bit[6]: REEN Bit[5:0]: PTM
P9:0xA0	CTRL_A0	0x01	RW	Bit[7:1]: Not used Bit[0]: CE
P9:0xA1	CTRL_A1	0xB7	RW	Bit[7:0]: tsup_cs
P9:0xA2	CTRL_A2	0x43	RW	Bit[7:0]: thp_cs
P9:0xA3	CTRL_A3	0x29	RW	Bit[7:0]: TSQ
P9:0xA4	CTRL_A4	0x1B	RW	Bit[7:0]: TRD

table 5-10 OTP registers (sheet 4 of 4)

address	register name	default value	R/W	description
P9:0xA5	CTRL_A5	0x15	RW	Bit[7:0]: Tgap
P9:0xA6	CTRL_A6	0xB5	RW	Bit[7:0]: tstr_sup
P9:0xA7	CTRL_A7	0xB9	RW	Bit[7:0]: TSTR
P9:0xA8	CTRL_A8	0x00	RW	Bit[7:5]: Not used Bit[4]: TE Bit[3]: ECC mode Bit[2]: DMA mode Bit[1]: TCS Bit[0]: TRS
P9:0xA9	CTRL_A9	0x00	RW	Bit[7:4]: Not used Bit[3:0]: otp_page
P9:0xAA~ P9:0xAC	RSVD	—	—	Reserved
P9:0xAD	CTRL_AD	0x00	RW	Bit[7:3]: Not used Bit[2:0]: otp_a_tmp2
P9:0xAE	RSVD	—	—	Reserved
P9:0xAF	CTRL_AF	0x06	RW	Bit[7:0]: t_blen_start
P9:0xB0	CTRL_B0	0x20	RW	Bit[7:0]: t_hvset
P9:0xB1	CTRL_B1	0x28	RW	Bit[7:0]: t_pctdd
P9:0xB2	CTRL_B2	0x50	RW	Bit[7:0]: t_blen_s
P9:0xB3	CTRL_B3	0x01	RW	Bit[7:0]: t_blen_h
P9:0xB4	CTRL_B4	0x01	RW	Bit[7:6]: Not used Bit[5:1]: t_wlen_del Bit[0]: Quick ip_cs_en
P9:0xB5~ P9:0xBF	RSVD	—	—	Reserved
P9:0xC0	CTRL_C0	0x00	RW	Bit[7:6]: dpc_start_adr_reg[7:6] Bit[5:0]: Not used
P9:0xC1	CTRL_C1	0x18	RW	Bit[7:6]: Not used Bit[5:0]: dpc_start_adr_reg[5:0]
P9:0xC2	CTRL_C2	0x00	RW	Bit[7:6]: dpc_end_adr_reg[7:6] Bit[5:0]: Not used
P9:0xC3	CTRL_C3	0xFF	RW	Bit[7:6]: Not used Bit[5:0]: dpc_end_adr_reg[5:0]

5.11 page selection [0xFD]

table 5-11 page selection registers

address	register name	default value	R/W	description
0xFD	PAGE_FLG	0x00	RW	Bit[7:4]: Reserved Bit[3:0]: Page select 0000: System control 0001: Sensor control page 1 0010: Sensor control page 2 0011: ISP 0100: Dynamic DPC 0101: BLC 0110: OTP DPC 0111: DCG combine 1001: OTP Others: Reserved

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6 operating specifications

6.1 absolute maximum ratings

table 6-1 absolute maximum ratings

parameter	absolute maximum rating ^a
ambient storage temperature	-40°C to +125°C
supply voltage (with respect to ground)	V_{DD-A} 4.5V
	V_{DD-D} 3V
	V_{DD-IO} 4.5V
all input/output voltages (with respect to ground)	-0.3V to $V_{DD-IO} + 1V$
I/O current on any input or output pin	± 200 mA
peak solder temperature (10 second dwell time)	245°C

- a. exceeding absolute maximum ratings shown above invalidates all AC and DC electrical specifications and may result in permanent damage to device. Exposure to absolute maximum rated conditions for extended periods may affect device reliability.

6.2 functional temperature

table 6-2 functional temperature

parameter	range
operating temperature (for applications up to 90 fps) ^a	-30°C to +85°C junction temperature
stable image temperature ^b	0°C to +60°C junction temperature

- a. sensor functions, but image quality may be noticeably different at temperatures outside of stable image range
b. image quality remains stable throughout this temperature range

6.3 DC characteristics

table 6-3 DC characteristics ($-30^{\circ}\text{C} < T_J < 85^{\circ}\text{C}$)^{ab}

symbol	parameter	min	typ	max	unit
supply					
V_{DD-A}	supply voltage (analog)	2.7	2.8	2.9	V
V_{DD-D}	supply voltage (digital core)	1.14	1.2	1.26	V
V_{DD-IO}	supply voltage (digital I/O)	1.7	1.8	1.9	V
I_{DD-A}	active (operating) current		31		mA
I_{DD-D}			57		mA
I_{DD-IO}			0.01		mA
$I_{DDS-XSHUTDOWN}$	standby current		10		μA
digital inputs (typical conditions: $DOVDD = 1.8\text{V}$)					
V_{IL}	input voltage LOW			0.54	V
V_{IH}	input voltage HIGH	1.26			V
C_{IN}	input capacitor			10	pF
digital outputs (standard loading 25 pF)					
V_{OH}	output voltage HIGH	1.62			V
V_{OL}	output voltage LOW			0.18	V
serial interface inputs					
V_{IL}^c	SCL and SDA			0.54	V
V_{IH}^c	SCL and SDA	1.26			V

- a. external clock is stopped during measurement
 b. standby current is based on room temperature
 c. based on $DOVDD = 1.8\text{V}$

6.4 timing characteristics

figure 6-1 reference clock input timing diagram

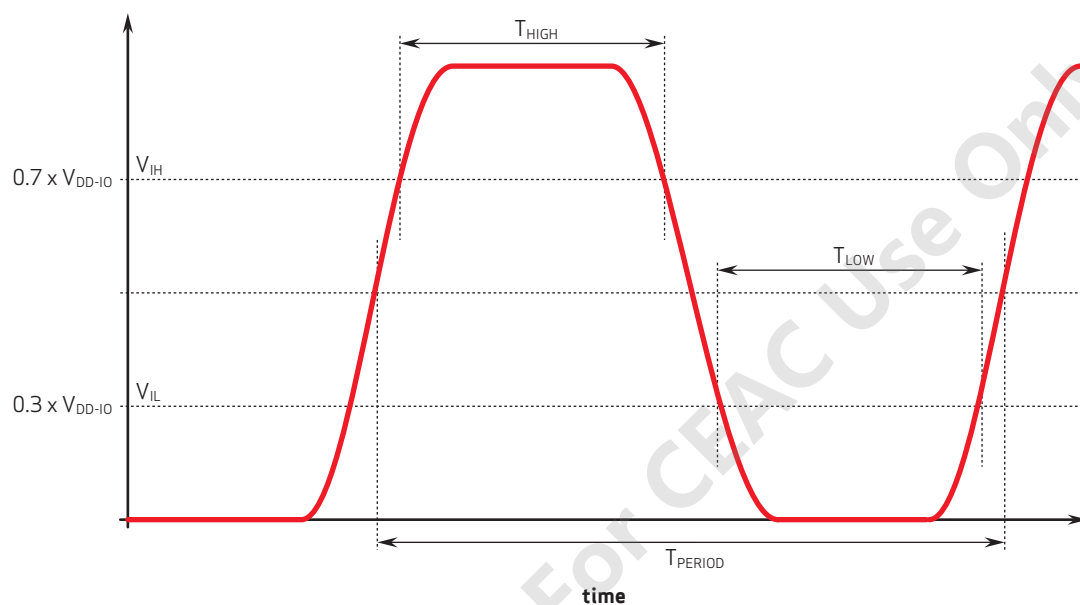


table 6-4 timing characteristics

symbol	parameter	min	typ	max	unit
oscillator and clock input					
f_{EXTCLK}	frequency (EXTCLK)	6	24	27	MHz
T_{PERIOD}	EXTCLK period	37.0	41.7	166.7	ns
V_{CLK}	EXTCLK amplitude	1.7	1.8	1.9	V
T_{HIGH}/T_{LOW}	EXTCLK duty cycle	$0.45 \times T_{PERIOD}$	$0.5 \times T_{PERIOD}$	$0.55 \times T_{PERIOD}$	ns

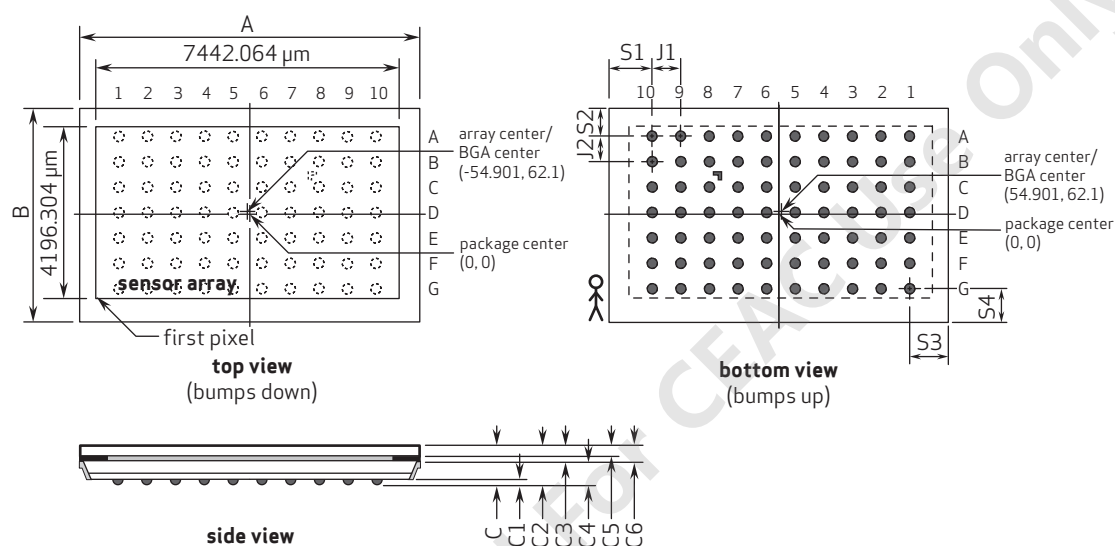
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7 mechanical specifications

7.1 physical specifications

figure 7-1 package specifications



note 1 this drawing is not to scale and is for reference only

note 2 all dimensions and coordinates are in microns unless otherwise specified

figure 7-2 final image output

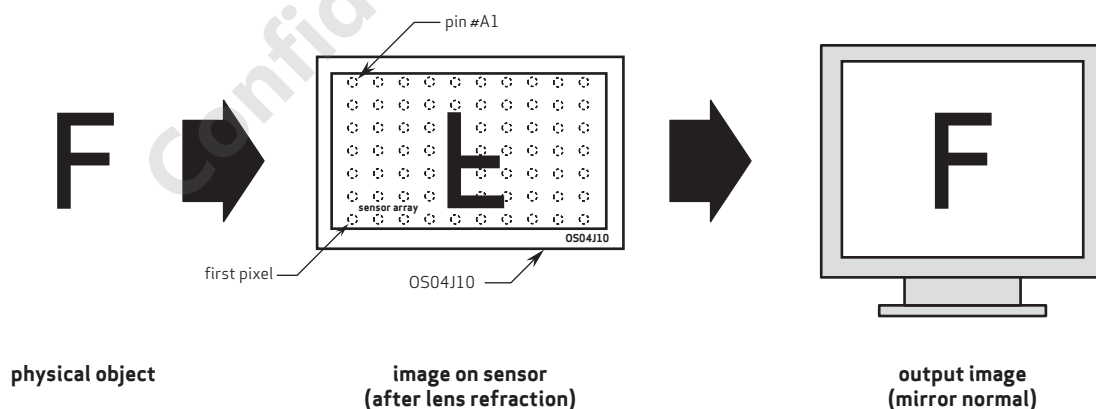
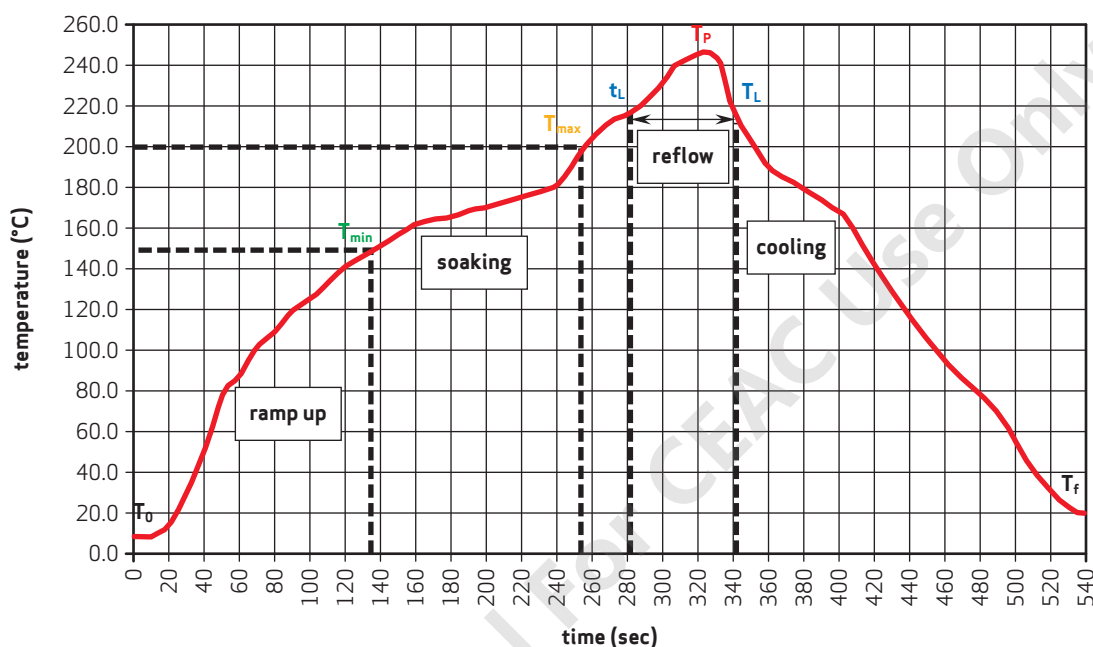


table 7-1 package dimensions

parameter	symbol	min	typ	max	unit
package body dimension x	A	8317.4	8342.4	8367.4	μm
package body dimension y	B	5189.9	5214.9	5239.9	μm
package height	C	710	770	830	μm
ball height	C1	100	130	160	μm
package body thickness	C2	605	640	675	μm
thickness from top glass surface to wafer	C3	430	445	460	μm
image plane height	C4	280	325	370	μm
glass thickness	C5	390	400	410	μm
air gap between sensor and glass	C6	41	45	49	μm
ball diameter	D	220	250	280	μm
total pin count	N		70 (12 NC)		
pins pitch x-axis	J1		700		μm
pins pitch y-axis	J2		620		μm
edge to pin center distance along x1	S1	1046.101	1076.101	1106.101	μm
edge to pin center distance along y1	S2	655.350	685.350	715.350	μm
edge to pin center distance along x2	S3	936.299	966.299	996.399	μm
edge to pin center distance along y2	S4	779.550	809.550	839.550	μm

7.2 IR reflow specifications

figure 7-3 IR reflow ramp rate requirements



note The OS04J10 uses a lead-free package.



note To reduce image artifacts from infrared light and to provide the best image quality, OmniVision recommends an IR cut filter.

table 7-2 reflow conditions^{ab}

zone	description	exposure
ramp up A (T_0 to T_{min})	heating from room temperature to 150°C	temperature slope $\leq 3^\circ\text{C}$ per second
soaking	heating from 150°C to 200°C	90 ~ 150 seconds
ramp up B (t_L to T_P)	heating from 217°C to 245°C	temperature slope $\leq 3^\circ\text{C}$ per second
peak temperature	maximum temperature in SMT	245°C $\pm 0/-5^\circ$ (duration max 30 sec)
reflow (t_L to T_L)	temperature higher than 217°C	30~120 seconds
ramp down A (T_P to T_L)	cooling down from 245°C to 217°C	temperature slope $\leq 3^\circ\text{C}$ per second
ramp down B (T_L to T_f)	cooling down from 217°C to room temperature	temperature slope $\leq 6^\circ\text{C}$ per second
T_0 to T_P	room temperature to peak temperature	≤ 8 minutes

a. maximum number of reflow cycles = 3

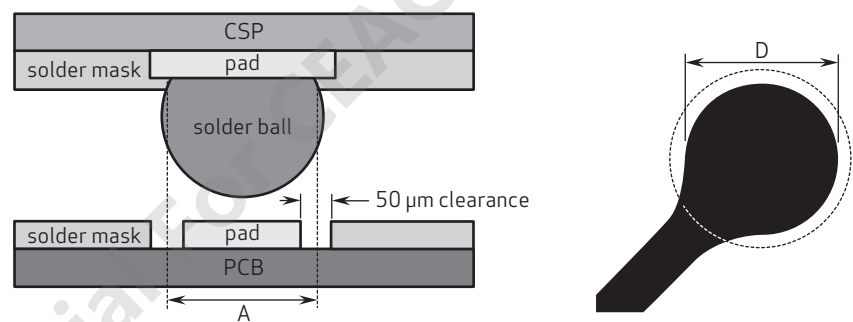
b. N2 gas reflow or control O2 gas PPM <500 as recommendation

7.3 PCB design recommendations

In general, OmniVision recommends circular non-solder mask defined (NSMD) PCB land pads, post-PCB fabrication size target of 90% of the package's ball pad opening, with a maximum allowable limit of 100% and minimum allowable limit of 80% of the package's ball pad opening, including manufacturing tolerance, with teardrop design (see **figure 7-4**) for trace to pad connections to minimize risk of cracking at the trace-pad interface.

- please refer to **table 7-3** for device specific PCB NSMD land pad size recommendations
- solder mask clearance around PCB NSMD land pads should be a minimum of 50 μm ; for applications requiring a clearance that is less than this recommended clearance, consult with PCB supplier
- high performance FR4 with high Tg and low CTE is recommended
- refer to IPC-7351B (*Generic Requirements for Surface Mount Design and Land Pattern Standard*) for additional details regarding any other PCB design considerations

figure 7-4 NSMD PCB land pad with tear drop example



note refer to **table 7-3** for A and D dimensions

table 7-3 recommended NSMD pad land pad sizes

device name	package type	package size	CSP/BGA ball pad opening size (A)	recommended PCB NSMD ball pad size (D)
OS04J10	CSP	8.3424 mm x 5.2149 mm	250 μm	225 μm

7.4 SMT recommendations

- stencil design should follow the IPC stencil guideline, IPC-7525B
- stencil: laser cut with electro-polishing
- solder material: SAC305 is recommended
- solder paste: prefer type 4 (20 μm to 38 μm) or finer solder sphere particle size, ROL 0
- SMT profile: refer to solder paste datasheet (provided by solder paste supplier) and recommended reflow profile

8 optical specifications

8.1 lens chief ray angle (CRA)

figure 8-1 chief ray angle (CRA)

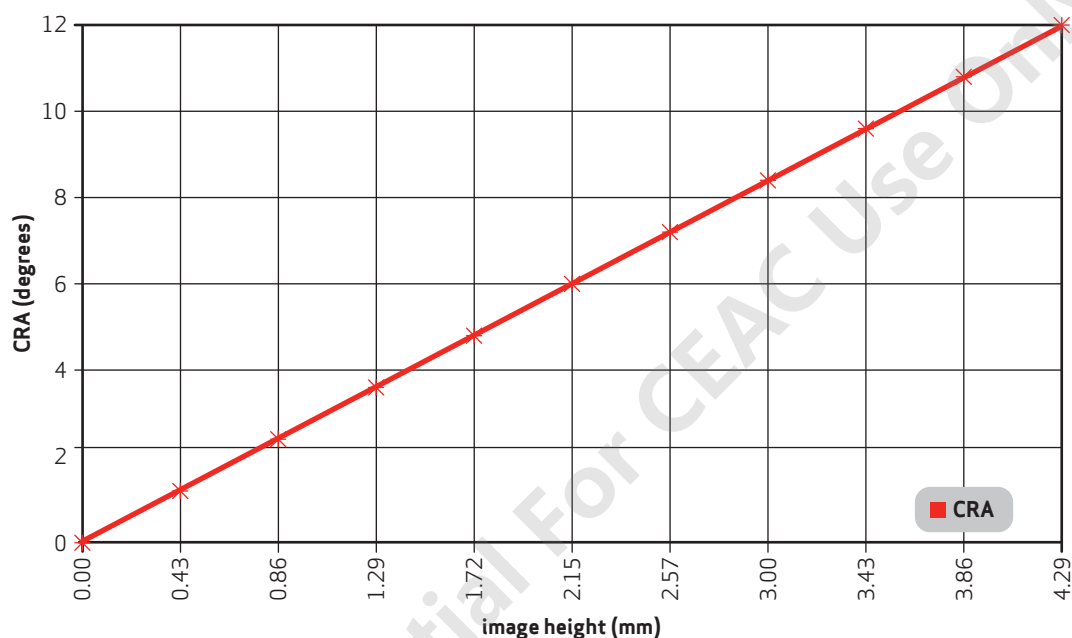


table 8-1 CRA versus image height plot

field (%)	image height (mm)	CRA (degrees)
0	0.00	0.00
10	0.43	1.20
20	0.86	2.40
30	1.29	3.60
40	1.72	4.80
50	2.15	6.00
60	2.57	7.20
70	3.00	8.40
80	3.43	9.60
90	3.86	10.80
100	4.29	12.00

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revision history

version 1.0 01.18.2023

- initial release

version 1.1 03.26.2024

- in key specifications, changed active power requirement to <200 mW and changed standby power requirement to <1 mA
- in section 2.5.2, changed sentence to "...by writing: 0xFC = 0x01"
- in section 2.6.2, changed first sentence of paragraph to "...power down {P0:0x09 = 0x07} through..."
- in section 4.4.1, changed fourth sentence of first paragraph to "Maximum exposure is VTS-8.", changed fifth sentence of paragraph to "...longer than VTS-8, the OS04J10..." and changed second sentence of second paragraph to "...registers are {P1:0x24} and the short frame ACG registers {P1:0xE7}."., changed third sentence and second paragraph to "...maximum AGC gain is 0xFF = 15.5x.", and changed third sentence of third paragraph to "...gain is 1/32: 0x40 is 1x digital gain, and the maximum digital gain is 0x7FE = 32x
- in section 4.4.2, changed first sentence of second paragraph to "...4 lines {P1:0x14 = 0x00 and P1:0x5 = 0x04} and the..."
- in table 5-3, changed description of register bits P1:0x1F[2:0] to dig_gain[9:7], changed description of register P1:0x20 to Bit[7:1]: dig_gain[6:0], Bit[0]: Not used, changed description of register bits P1:0xE4[2:0] to dig_gain_s[9:7], changed description of register P1:0xE5 to Bit[7:1]: dig_gain_s[6:0], Bit[0]: Not used, changed register name, default value, R/W, and description of register P1:0xE6 to NOT USED, -, -, and Not used, respectively
- in section 5.7, changed title to "BLC [P5:0x00 ~ P5:0xDB]"
- in table 5-7, changed register name, default value, R/W, and description of registers P1:0xBB to RSVD, -, -, and Reserved, respectively, changed register name, default value, R/W, and description of registers P1:0xBC to RSVD, -, -, and Reserved, respectively, changed register name, default value, R/W, and description of registers P1:0xBB to RSVD, -, -, and Reserved, respectively, and changed register name, default value, R/W, and description of registers P5:0xC3~P5:0xDA to RSVD, -, -, and Reserved, respectively
- in table 5-11, changed binary value selection 0101 of register bits 0xFD[3:0] to BLC
- in table 6-3, changed typ value for I_{DD-A} to 31 mA, changed typ value for I_{DD-IO} to 0.01 mA, changed typ value for I_{DD-D} to 57 mA, changed typ value for I_{DDS-XSHUTDOWN} to 10 µA

version 1.2 11.21.2024

- in ordering information, added ordering part number OS04J10-A70A-001B-Z
- in key specifications, changed sensitivity to 31661 e⁻/Lux-sec, changed max S/N ratio to 35.4 dB @ HCG, 42.2 dB @ LCG, changed dynamic range to 66.2 dB @ 1x, 77.6 dB @ 15.5x, and changed pixel size to 2.898 µm x 2.898 µm
- in section 2.11.1, completely replaced figure 2-7
- in section 2.11.2, completely replaced figure 2-8 and table 2-8

- in chapter 4, added section 4.8
- in figure 7-1, removed a and b dimensions in side view
- in table 7-1, removed last two rows
- in table 7-2, changed ramp down B (T_L to T_f) exposure to temperature slope $\leq 6^\circ\text{C}$ per second
- in chapter 7, added section 7.3

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OMNIVISION®

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OmniVision Technologies, Inc.

UNITED STATES

4275 Burton Drive
Santa Clara, CA 95054
tel: +1 408 567 3000
Chicago +1 847 508 0217
Detroit +1 248 919 6900
Optics Division +1 303 449 5593

UNITED KINGDOM

Hants +44 1252 761601

GERMANY

Munich +49 89 6381 9988

CHINA

Shanghai +86 21 6175 9888
+86 21 6154 8000
Beijing +86 10 8642 4200
Shenzhen +86 755 8435 9733
Wuhan +86 27 5975 7800
Xian +86 29 8885 2537

INDIA

Bangalore +91 80 4112 8966

KOREA

Seoul +82 2 3472 3769

JAPAN

Yokohama +81 45 478 7977
Kyoto +81 75 708 5352

NORWAY

Oslo +47 2295 8100

SINGAPORE +65 6933 1933

TAIWAN

Taipei +886 2 2657 9800
Hsin-chu +886 3 5656688